Linear Algebra

Jaysen Tsao

1 Linear Equations

1.1 Systems of Linear Equations

Linear Function

A function f is **linear** in the variables $x_1, x_2, \ldots, x_n$ if it can be written in the form:

$$f\left(x_1, x_2, \ldots, x_n, \ldots\right) = a_1 x_1 + a_2 x_2 + \ldots + a_n x_n$$

where $a_1, a_2, \ldots, a_n$ are constants with respect to $x_1, x_2, \ldots, x_n$.

- To put it simply, a function is **linear** in the variables $x_1, x_2, \ldots, x_n$ if it can be written as the sum of constant multiples of $x_1, x_2, \ldots, x_n$. These constant multiples are called the **coefficients**.
- For example, the function $f(x, y) = 2xy^2$ is linear in x but not linear in y.

Linear Equation

A **linear equation** in the variables $x_1, x_2, \ldots, x_n$ can be written in the form:

$$f\left(x_1, x_2, \ldots, x_n, \ldots\right) = b$$

where f is linear in $x_1, x_2, \ldots, x_n$ and b is constant with respect to $x_1, x_2, \ldots, x_n$.

- A **linear equation** in the variables $x_1, x_2, \ldots, x_n$ can be written in the form:

  $$a_1 x_1 + a_2 x_2 + \ldots + a_n x_n = b$$

  where a_i is the coefficient of the variable x_i for all $1 \leq i \leq n$, and b is a constant.
- For example, the equation $2x + 5y = 6z^2$ is linear in x and y but not linear in z.

System of Linear Equations

A **system of linear equations** (or **linear system**) is a collection of one or more linear equations involving the same set of variables.

A **solution** of a linear system in the variables $x_1, x_2, \ldots, x_n$ is an ordered n-tuple $\left(s_1, s_2, \ldots, s_n\right)$ that, when substituted for $\left(x_1, x_2, \ldots, x_n\right)$ respectively, makes each equation in the system a true statement.

- The **solution set** of a system of equations is the set of all possible solutions for that system.
  - Two systems are **equivalent** if they have the same solution set.
- A linear system can have one solution, no solution, or infinitely many solutions.
  - A linear system is **consistent** if it has at least one solution (i.e., one or infinite)
  - A linear system is **inconsistent** if it has no solutions.

Example: Solving and Classifying a Linear System

Find the solution set to the following system of equations, and classify the system as consistent or inconsistent:

$$\begin{aligned} x_1 - 2x_2 &= -1 \\ -x_1 + 3x_2 &= 3 \end{aligned}$$

Adding the two equations together, we get $x_2 = 2$, which implies $x_1 - 2(2) = -1 \implies x_1 = 3$.

Thus, one solution to the system is $(3, 2)$. Since it is the only solution, the solution set is $\{(3, 2)\}$, and the system is consistent.

Example: Textbook Exercise 1.1.P4

For what values of h and k make the following system consistent?

$$\begin{aligned} 2x_1 - x_2 &= h \\ -6x_1 + 3x_2 &= k \end{aligned}$$

We can multiply the first equation by 3 and add it to the second equation:

$$0x_1 + 0x_2 = k + 3h \implies 0 = k + 3h$$

For the system to be consistent, we need $k + 3h = 0 \implies k = -3h$.

Otherwise, if $k + 3h \neq 0$, the system is inconsistent because there would be no solutions (i.e., a contradiction).

Introduction to Matrices

- A **matrix** is a collection of numbers arranged into a rectangular array of rows and columns.
  - A matrix with m rows and n columns is called an $m \times n$ matrix.
  - If A is a matrix, then a_{ij} denotes the entry in the i-th row and j-th column of A.
- To distinguish between scalars, vectors, and matrices, multiple notations are used:
  - Scalars are lowercase and italicized, e.g., a, b, c.
  - Vectors can either be:
    - lowercase and bolded: $\mathbf{v}, \mathbf{w}$, or
    - lowercase with an arrow on top: $\mathbf{v}, \mathbf{w}$.
    - unit vectors are sometimes denoted with a hat: $\hat{\imath}, \hat{\jmath}, \hat{k}$.
  - Matrices are uppercase and (usually) bolded, e.g., $\mathbf{A}, \mathbf{B}, \mathbf{C}$.
    - sometimes they are simply italicized: A, B, C (as in PDP^{-1})
- A vector $\langle v_1, v_2, v_3, \dots \rangle$ can be represented as a matrix (or made “compatible” with matrices) by writing it as either a **column matrix** (or **column vector**): $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ \vdots \end{pmatrix}$

  or a **row matrix** (or **row vector**): $\mathbf{v} = \begin{pmatrix} v_1 & v_2 & v_3 & \dots \end{pmatrix}$

- The **coefficient matrix** of a linear system is the matrix consisting of the coefficients of the variables in the system. For example, the system:

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ 2x_2 - 8x_3 &= 8 \\ 5x_1 - 5x_3 &= 10 \end{aligned}$$

can be represented by the coefficient matrix $\begin{pmatrix} 1 & -2 & 1 \\ 0 & 2 & -8 \\ 5 & 0 & -5 \end{pmatrix}$.

- The **augmented matrix** of a linear system is the matrix consisting of the coefficient matrix with an additional column that contains the constants from the right-hand side of each equation.

  For example, the augmented matrix for the above system is $\begin{pmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 5 & 0 & -5 & 10 \end{pmatrix}$.

Solving Linear Systems

One strategy to solve a linear system is to perform **elementary row operations** on the associated augmented matrix, eliminating variables one at a time until we arrive at a trivial form.

An example of a trivial form is the **triangular form**, where all entries below the main diagonal are zero. Below is a template for a 3×4 triangular matrix:

$$\begin{pmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \end{pmatrix}$$

The main diagonal is highlighted in blue.

Elementary Row Operations

There are three **elementary row operations** which operate on individual rows of a matrix (which can correspond to equations in a linear system):

1. *Swap* two rows $R_i \leftrightarrow R_j$
2. *Multiply* by a nonzero scalar: $R_i \to kR_i$ where $k \neq 0$
3. *Add* a multiple of one row to another: $R_i \to R_i + kR_j$

Note: Operation 3 is sometimes $R_i \to aR_i + bR_j$

- Row operations can be applied to any matrix.
- Two matrices are **row equivalent** if there exists a sequence of elementary row operations that transforms one matrix into the other.
  - If the augmented matrices of two linear systems are row equivalent, then the two systems are equivalent (i.e., have the same solution set).
- Row operations are *reversible*, and the following lists inverse operations:
  - $R_i \leftrightarrow R_j$ is its own inverse.
  - $R_i \to kR_i$ is inverted by $R_i \to \frac{1}{k}R_i$.

- $R_i \to R_i + kR_j$ is inverted by $R_i \to R_i - kR_j$.

Example: Solving a Linear System

Solve the following system of equations:

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ 2x_2 - 8x_3 &= 8 \\ 5x_1 - 5x_3 &= 10 \end{aligned}$$

We can use the augmented matrix to represent the system, and label the rows:

$$\begin{pmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 5 & 0 & -5 & 10 \end{pmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}$$

Start with the equation with the most leading zeros, which is R_2. We can use this to eliminate x_2 from R_1 by adding R_2 to R_1 ($R_1 \to R_1 + R_2$):

$$\begin{pmatrix} 1 & 0 & -7 & 8 \\ 0 & 2 & -8 & 8 \\ 5 & 0 & -5 & 10 \end{pmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}$$

Notice that only R_1 changed, and that we targeted R_1 because it had the most nonzero entries.

Next, we can use R_1 to eliminate x_1 from R_3 by performing the operation $R_3 \to R_3 - 5R_1$:

$$\begin{pmatrix} 1 & 0 & -7 & 8 \\ 0 & 2 & -8 & 8 \\ 0 & 0 & 30 & -30 \end{pmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}$$

Finally, we can use R_3 to eliminate x_3 from R_2 by performing the operation $R_2 \to R_2 + \frac{4}{15}R_3$:

$$\begin{pmatrix} 1 & 0 & -7 & 8 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 30 & -30 \end{pmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}$$

We have achieved triangular form, so we can start back-substitution. Here, we see the final matrix corresponds to the system of equations:

$$\begin{aligned} x_1 - 7x_3 &= 8 \\ 2x_2 &= 0 \implies x_2 = 0 \\ 30x_3 &= -30 \implies x_3 = -1 \end{aligned}$$

Finally, $x_1 - 7(-1) = 8 \implies x_1 + 7 = 8 \implies x_1 = 1$. Thus, the solution is $(1, 0, -1)$.

1.2 Row Reduction & Echelon Forms

Echelon Form

A rectangular matrix is in **echelon form** or **row echelon form** (REF) if it is:

1. *Separated*: nonzero rows are above any rows of all zeros.
2. *Ordered*: the leading entry (or **pivot**) of each nonzero row is to the right of the leading entry of the previous row.
3. *Normalized Below*: all entries below a leading entry are zeros.

The leading entry (or **pivot**) of a nonzero row is the first nonzero entry from the left in that row.

A matrix is in **reduced row echelon form** (or **row reduced echelon form**, RREF) if it is in echelon form and also meets the following additional properties:

1. *Leading 1s*: the leading entry in each nonzero row is 1.
2. *Normalized Everywhere*: all entries above and below a leading 1 are zeros.

- A matrix in echelon form must be in triangular form, but the converse is not necessarily true.
- A matrix in echelon form is called an **echelon matrix**, and a matrix in RREF is called a **reduced echelon matrix**.

Existence and Uniqueness of Reduced Row Echelon Form

Every matrix is row equivalent to one and only one reduced echelon matrix.

*In other words, each matrix has **exactly one** reduced row echelon form.*

Pivots and Free Variables

- The **pivot** positions of a matrix are the locations of the leading entries in its nonzero rows.
  - The columns that contain pivot positions are called **pivot columns**.
  - The number of pivot positions is called the **rank** of the matrix.
- The columns that do not contain pivot positions are called **free columns**.
  - The variables corresponding to free columns are called **free variables**.
    As their name suggests, these variables can take on any value in the solution set.
  - The other variables (variables in pivot columns) are called **basic variables**.
- Basic variables may be expressed in terms of the free variables in a linear system. This way of expressing the solution set is called the **parametric description** (or form) of the solution set.
  - The following are two equivalent parametric descriptions:

$$\left\{ \left(x_1, x_1 + 2\right) \,\middle|\, x_1 \in \mathbb{R} \right\} \equiv \begin{cases} x_1 \text{ is free} \\ x_2 = x_1 + 2 \end{cases}$$

Existence Theorem

Let A be the augmented matrix of a linear system. If the rightmost column of A is not a pivot column, then the linear system must be consistent (and there must exist at least one solution).

- The Existence Theorem is intuitive to understand. The only way to have the rightmost column be a pivot is to have a row in the form $(0, 0, \ldots, b)$ with $b \neq 0$. This corresponds to the equation $0 = b$, which is a contradiction and thus makes the system inconsistent.

Uniqueness Theorem

Let A be the augmented matrix of a *consistent* linear system. If every variable in the system is a basic variable (i.e., there are no free variables), then the linear system has a unique solution.

The Row Reduction Algorithm

Let's consider the following matrix:

$$\begin{pmatrix} 1 & 2 & -1 & -4 \\ 2 & 3 & 3 & 3 \\ 4 & 5 & 1 & 6 \end{pmatrix}$$

To reduce this matrix into **echelon form** (ref), we can follow these steps (the **forward phase**):

1. Choose the leftmost column as the **pivot column**. Our goal with a pivot column is to turn it into a column with a pivot position in the first row. That is, we want to have a leading entry in the first row and zeros below it.

$$\begin{pmatrix} \color{red}{1} & 2 & -1 & -4 \\ \color{blue}{2} & 3 & 3 & 3 \\ \color{blue}{4} & 5 & 1 & 6 \end{pmatrix}$$

The pivot column is highlighted in blue, and the pivot position is highlighted in red.
Only focusing on the blue column, we want to use row operations to make everything below the red entry equal to zero.

2. If necessary, swap rows to move a nonzero entry into the pivot position (first row of the pivot column). In this case, the first entry is already nonzero, so we can skip this step.

3. Use row operations to create zeros below the pivot position. We can eliminate the entries below the pivot by performing the following row operations:

$$R_2 \to R_2 - 2R_1 \qquad \begin{pmatrix} \color{red}{1} & 2 & -1 & -4 \\ \color{blue}{0} & -1 & 5 & 11 \\ \color{blue}{4} & 5 & 1 & 6 \end{pmatrix}$$

$$R_3 \to R_3 - 4R_1 \qquad \begin{pmatrix} \color{red}{1} & 2 & -1 & -4 \\ \color{blue}{0} & -1 & 5 & 11 \\ \color{blue}{0} & -3 & 5 & 22 \end{pmatrix}$$

4. Now, consider the pivot column as the next column to the right (that contains a nonzero entry). The row of our pivot position will also move down one row.

   In this case, we can use the second column as the next pivot column:

$$\begin{pmatrix} 1 & 2 & -1 & -4 \\ 0 & -1 & 5 & 11 \\ 0 & -3 & 5 & 22 \end{pmatrix}$$

   Performing row operations to create the zero below the pivot position:

$$R_3 \to R_3 - 3R_2 \qquad \begin{pmatrix} 1 & 2 & -1 & -4 \\ 0 & -1 & 5 & 11 \\ 0 & 0 & -10 & -11 \end{pmatrix}$$

5. Finally, we can use the third column as the next pivot column:

$$\begin{pmatrix} 1 & 2 & -1 & -4 \\ 0 & -1 & 5 & 11 \\ 0 & 0 & -10 & -11 \end{pmatrix}$$

   Since there are no entries below the pivot position, we have achieved echelon form (REF). This process was called **Gaussian Elimination**.

To convert the matrix into **reduced row echelon form** (RREF), we can follow these steps (the **backward phase**):

1. Starting from the *last* pivot column, use row operations to create zeros *above* the pivot position. Starting with the third column:

$$R_2 \to R_2 + \frac{1}{2}R_3 \qquad \begin{pmatrix} 1 & 2 & -1 & -4 \\ 0 & -1 & 0 & 5.5 \\ 0 & 0 & -10 & -11 \end{pmatrix}$$

$$R_1 \to R_1 - \frac{1}{10}R_3 \qquad \begin{pmatrix} 1 & 2 & 0 & -2.9 \\ 0 & -1 & 0 & 5.5 \\ 0 & 0 & -10 & -11 \end{pmatrix}$$

2. If the pivot is not 1, scale accordingly:

$$R_3 \to -\frac{1}{10}R_3 \qquad \begin{pmatrix} 1 & 2 & 0 & -2.9 \\ 0 & -1 & 0 & 5.5 \\ 0 & 0 & 1 & 1.1 \end{pmatrix}$$

3. Repeat steps 1 and 2 for the remaining pivot columns, moving leftward:

$$R_1 \to R_1 + 2R_2 \qquad \begin{pmatrix} 1 & 0 & 0 & 8.1 \\ 0 & -1 & 0 & 5.5 \\ 0 & 0 & 1 & 1.1 \end{pmatrix}$$

$$R_2 \to -1R_2 \qquad \begin{pmatrix} 1 & 0 & 0 & 8.1 \\ 0 & 1 & 0 & -5.5 \\ 0 & 0 & 1 & 1.1 \end{pmatrix}$$

$$\text{left column is good} \qquad \begin{pmatrix} 1 & 0 & 0 & 8.1 \\ 0 & 1 & 0 & -5.5 \\ 0 & 0 & 1 & 1.1 \end{pmatrix}$$

This is now in reduced row echelon form (RREF), and we can read off the solution directly:

$$\left(x_1, x_2, x_3\right) = (8.1, -5.5, 1.1).$$

This backward process is called **Jordan Elimination**, and the entire process of converting a matrix into RREF is called **Gauss-Jordan Elimination**.

1.3 Vector & Matrix Equations

- A **column vector** is a matrix with a single column.
  - The set of all n-dimensional column vectors with real entries is denoted by $\mathbb{R}^n$.
- Vectors in $\mathbb{R}^2$ can be thought of as points or arrows in the Cartesian xy-plane, while vectors in $\mathbb{R}^3$ can be thought of as points or arrows in three-dimensional space.

Algebraic Properties of Vector Spaces

1. Commutative under addition: $\mathbf{v}_1 + \mathbf{v}_2 = \mathbf{v}_2 + \mathbf{v}_1$.
2. Associative under addition: $\left(\mathbf{v}_1 + \mathbf{v}_2\right) + \mathbf{v}_3 = \mathbf{v}_1 + \left(\mathbf{v}_2 + \mathbf{v}_3\right)$.
3. Additive identity: There exists a zero vector $\mathbf{0}$ such that $\mathbf{v} + \mathbf{0} = \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$.
4. Additive inverses: For every $\mathbf{v} \in \mathbb{R}^n$, there exists a vector $-\mathbf{v}$ such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$.
5. Distributive properties: $c\left(\mathbf{v}_1 + \mathbf{v}_2\right) = c\mathbf{v}_1 + c\mathbf{v}_2$ and $\left(c_1 + c_2\right)\mathbf{v} = c_1\mathbf{v} + c_2\mathbf{v}$.
6. Compatibility with scalar multiplication: $c_1\left(c_2\mathbf{v}\right) = \left(c_1c_2\right)\mathbf{v}$.
7. Multiplicative identity: $1\mathbf{v} = \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$.

- A **vector space** is *any* set of vectors that satisfies all the algebraic properties listed above. For example, $\mathbb{R}^n$ all subsets of $\mathbb{R}^n$ are vector spaces (for any n).
  - A **subspace** is a vector space that is contained within another vector space
    i.e., a vector space that is a subset of another vector space.
- A **linear combination** of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is an expression of the form:

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n$$

  where $c_1, c_2, \dots, c_n$ are scalars. These scalars are called the **weights** of the linear combination.
- The **span** of a set of vectors is the set of all possible linear combinations of those vectors. The span of a set of vectors in $\mathbb{R}^n$ is always a subspace of $\mathbb{R}^n$.

Span of a Set of Vectors

Let $V = \left\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\right\} \subseteq \mathbb{R}^n$. The **span** of V, denoted $\operatorname{span}(V)$ or $\operatorname{span}\left\{\mathbf{v_1}, \mathbf{v}_2, \dots, \mathbf{v}_n\right\}$, is the set of all linear combinations of the vectors in V:

$$\operatorname{span}(V) = \left\{\mathbf{v} \in \mathbb{R}^n \mid \mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n \text{ for } c_1, c_2, \dots, c_n \in \mathbb{R}\right\}.$$

It holds that $\operatorname{span}(V)$ is a subspace of $\mathbb{R}^n$.

- A matrix can be written in terms of column vectors representing its columns as follows:

$$A = \begin{pmatrix}\mathbf{a}_1 & \mathbf{a}_2 & \dots & \mathbf{a}_n\end{pmatrix}$$

  where $\mathbf{a}_i$ is the i^{th} column of $\mathbf{A}$ for $i = 1, 2, \dots, n$.
- A **vector equation** in the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is an equation of the form:

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n = \mathbf{b}$$

where $c_1, c_2, \ldots, c_n$ are scalars and $\mathbf{b}$ is a vector. It has the same solution set as the linear system whose augmented matrix is $(\mathbf{v}_1\ \ \mathbf{v}_2\ \ \ldots\ \ \mathbf{v}_n\ \ \mathbf{b})$.

Matrix Equations

Let A be the $m \times n$ matrix $\begin{pmatrix}\mathbf{a}_1 & \mathbf{a}_2 & \ldots & \mathbf{a}_n\end{pmatrix}$, such that $\mathbf{a}_i$ is the i^{th} column of A. Let $\mathbf{x}$ be the vector $\begin{pmatrix}x_1\\x_2\\\vdots\\x_n\end{pmatrix}$.

Then the **matrix product** $A\mathbf{x}$ is defined as the linear combination of the columns of A with weights given by the entries of $\mathbf{x}$:

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \ldots + x_n\mathbf{a}_n.$$

Realize that $A\mathbf{x}$ is a vector in $\mathbb{R}^m$. We can constrain this vector to equal another vector $\mathbf{b}$ in $\mathbb{R}^m$ to get a **matrix equation**:

$$A\mathbf{x} = \mathbf{b}.$$

This matrix equation has the same solution set as the linear system whose augmented matrix is $\begin{pmatrix}\mathbf{a}_1 & \mathbf{a}_2 & \ldots & \mathbf{a}_n & \mathbf{b}\end{pmatrix}$. In other words, a matrix equation is another way of representing a linear system.

So, we have four ways to represent a linear system: as a system of a linear equations, as an augmented matrix, as a vector equation, and as a matrix equation.

The coefficient matrix can tell us a few things:

- if all columns are pivot columns, there are no free variables. if the system is also consistent, then it has a unique solution. (i.e., if all columns are pivot columns, the system has at most one solution)
  - if there are free variables, then the system has infinitely many solutions (if it is consistent).
- if all rows are pivot rows, then the system MUST be consistent (at least one solution)

Transition Matrices

A **transition matrix**, **migration matrix**, or **Markov chain** P describes the probabilities of each state transitioning to every other state in a system. They have the following properties:
- All entries are nonnegative: $P_{ij} \geq 0$ for all i, j.
- The entries in each column sum to 1.

Each column represents a probability distribution of one state in the system transitioning to other states. For example, if $P_{ij} = p$, then there is a p probability of transitioning from state j to state i.

When multiplied by a vector $\mathbf{v}$ representing the distribution of states at a given time, we get the distribution of those states at the next time step. Thus, transition matrices can be used to model the evolution of systems over time:

$$\mathbf{v}_{\text{next}} = P\mathbf{v}$$
$$\mathbf{v}_{\text{after } k \text{ steps}} = P^k\mathbf{v}.$$

Some transition matrices have a special property that when multiplied by a vector representing the distribution of states, the distribution does not change. In other words, there may exist a vector $\mathbf{v}$ such that $P\mathbf{v} = \mathbf{v}$. Such a vector is called a **steady state** or **equilibrium** of the system.

We can find a potential steady-state for the system by solving the equation $P\mathbf{v} = \mathbf{v}$, which is equivalent to solving the homogeneous system $(P - I)\mathbf{v} = \mathbf{0}$, where I is the identity matrix.

Example: Applying a Transition Matrix

18000 students live on campus. Each day, each student either brings their lunch or goes to the cafeteria to buy lunch.

1. On day 0, 9000 brought lunch and 9000 went to the cafeteria.
2. About 80% of students who brought lunch the previous day bring lunch again the next day.
3. About 60% of students who went to the cafeteria the previous day go to the cafeteria again the next day.

How many students will bring lunch and how many will go to the cafeteria on day 1? On day 2? On day k? After an infinite number of days?

We can represent the distribution of students on day k as a vector $\mathbf{v}_k = \begin{pmatrix} b_k \\ c_k \end{pmatrix}$.
We are given $\mathbf{v}_0 = \begin{pmatrix} 9000 \\ 9000 \end{pmatrix}$.

b_k is the number of students who bring lunch on day k and c_k is the number of students who go to the cafeteria on day k.

We can set up a grid to determine the transition probabilities:

1.4 Solution Sets of Linear Systems

- A system of linear equations is **homogeneous** if it can be written in the form $A\mathbf{x} = \mathbf{0}$, where A is the coefficient matrix and $\mathbf{0}$ is the zero vector.
  - A homogeneous system is *always* consistent because the **trivial solution** $\mathbf{x} = \mathbf{0}$ always satisfies the equation.
- For homogeneous systems, we are often interested in finding the **nontrivial solutions**.

Existence of a Nontrivial Solution

Let A be the coefficient matrix of a homogeneous linear system. The system has a nontrivial solution if and only if the equation $A\mathbf{x} = \mathbf{0}$ has at least one free variable.

1. Let $\mathbf{x} = \mathbf{x}_h$ solve $A\mathbf{x} = \mathbf{0}$. Then, for any constant $c \in \mathbb{R}$, $\mathbf{x} = c\mathbf{x}_h$ is also a solution to $A\mathbf{x} = \mathbf{0}$.
2. If $\mathbf{x} = \mathbf{p}$ solves $A\mathbf{x} = \mathbf{b}$, then the solution set of $A\mathbf{x} = \mathbf{b}$ is given by:

$$\left\{\mathbf{x} = \mathbf{p} + \mathbf{x}_h \mid A\mathbf{x}_h = \mathbf{0}\right\}.$$

   In other words, the solution set of a nonhomogeneous linear system with solution $\mathbf{x} = \mathbf{p}$ is a translation of the solution set of the corresponding homogeneous linear system by $\mathbf{p}$. This is often referred to as the **Translation Theorem**.

Proof: Translation Theorem

Say $\mathbf{x} = \mathbf{x}_h$ is a solution for $A\mathbf{x} = \mathbf{0}$ and $\mathbf{x} = \mathbf{p}$ is a solution for $A\mathbf{x} = \mathbf{b}$. Then $\mathbf{x} = \mathbf{p} + \mathbf{x}_h$ is also a solution for $A\mathbf{x} = \mathbf{b}$ because:

$$A\mathbf{x} = A\left(\mathbf{p} + \mathbf{x}_h\right) = A\mathbf{p} + A\mathbf{x}_h = \mathbf{b} + \mathbf{0} = \mathbf{b}.$$

Uniqueness of Solutions

If the *only* solution to $A\mathbf{x} = \mathbf{0}$ is the trivial solution $\mathbf{x}_h = \mathbf{0}$, then $A\mathbf{x} = \mathbf{b}$ has exactly one solution.

Summary

- Homogeneous systems $A\mathbf{x} = \mathbf{0}$ are always consistent, they have at least one *trivial* solution $\mathbf{x} = \mathbf{0}$.
  - A homogeneous system has a *nontrivial* solution if and only if it has at least one free variable, in which case it will have *infinitely many* solutions.
- Any system $A\mathbf{x} = \mathbf{b}$ with a known solution $\mathbf{x} = \mathbf{p}$ will *also* have the solution $\mathbf{x} = \mathbf{x}_h + \mathbf{p}$ where $\mathbf{x}_h$ is a solution to the corresponding homogeneous system $A\mathbf{x} = \mathbf{0}$.
  - If the homogeneous system only has the trivial solution, then the nonhomogeneous system will have a unique solution.
  - Otherwise, if the homogeneous system has a nontrivial solution, then the nonhomogeneous system will have infinitely many solutions.

1.5 Linear Independence

- Two vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ in $\mathbb{R}^n$ are **linearly independent** if the only solution to the equation:

  $$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = \mathbf{0}$$

  is the trivial solution $c_1 = 0$ and $c_2 = 0$.
- A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ in $\mathbb{R}^n$ is **linearly independent** if the only solution to the equation:

  $$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n = \mathbf{0}$$

  is the trivial solution $c_1 = c_2 = \dots = c_n = 0$.
  - In other words, a set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is linearly independent if $\operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} = \mathbb{R}^n$.
  - A set of one vector $\{\mathbf{v}\}$ is linearly independent if and only if $\mathbf{v} \neq \mathbf{0}$.
  - Two vectors are linearly independent if and only if one is *any* scalar multiple of the other.
- Conversely, a set of vectors is **linearly dependent** if there exists a nontrivial solution to the equation $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n = \mathbf{0}$.
  - If a set of vectors V is linearly dependent, then at least one vector in V is a linear combination of the others. Formally, $\exists \mathbf{v} \in V$ such that $\mathbf{v} \in \operatorname{span}(V \setminus \{\mathbf{v}\})$.

    In other words, if we can remove a vector from V such that $\operatorname{span}(V)$ remains the same, then (the original) V is linearly dependent.
- The columns of a matrix A are linearly independent iff the only solution to $A\mathbf{x} = \mathbf{0}$ is the trivial solution $\mathbf{x} = \mathbf{0}$.
  - The only way a homogeneous system has a nontrivial solution is if it has infinite solutions, which only happens if there is at least one free variable.

    Thus, the columns of A are linearly independent iff there are no free variables (i.e., every column is a pivot column).
- If a set of vectors contains the zero vector, then the set is linearly dependent. This is because we can have a nontrivial solution by setting the weight of the zero vector to be any nonzero scalar and setting the weights of all other vectors to be zero.
- If a set of vectors contains more vectors than the number of entries in each vector, then the set is linearly dependent. This is because there must be at least one free variable.
  - That is, if a vector space $V \subseteq \mathbb{R}^n$ contains more than n vectors, then V is linearly dependent.

Proof: If $V \subseteq \mathbb{R}^n$ and $|V| > n$, then V is linearly dependent

Let $V = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m\} \subseteq \mathbb{R}^n$ where $m > n$, so $|V| = m > n$. Consider the equation:

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_m\mathbf{v}_m = \mathbf{0}.$$

This is a linear system with m variables and n equations, with its coefficient matrix having m columns and n rows. Each row can only have at most one pivot, so there can be at most n pivots. Since $m > n$, there are more variables than pivots, so there must be at least one variable without a pivot, i.e. a free variable. Thus, there is a nontrivial solution to the equation, so V is linearly dependent.

1.6 Linear Transformations

- A **transformation** is a function from one vector space to another. $T : \mathbb{R}^n \to \mathbb{R}^m$
  - Here, $\mathbb{R}^n$ is called the **domain** of T, and $\mathbb{R}^m$ is called the **codomain** of T.
  - The transformation of a vector $\mathbf{v}$ by T is denoted by $T(\mathbf{v})$, and is called the **image** of $\mathbf{v}$ under T.
  - The set of all images of vectors in the domain of T is called the **range** of T.
  - If $T(\mathbf{u}) = \mathbf{v}$, then we can use the shorthand notation $\mathbf{u} \mapsto \mathbf{v}$.
- A transformation T is **linear** if two properties are satisfied:
  - $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all vectors $\mathbf{u}$ and $\mathbf{v}$ in the domain of T.
  - $T(c\mathbf{v}) = cT(\mathbf{v})$ for all vectors $\mathbf{v}$ in the domain of T and all scalars c.
- A linear transformation T thus have the following properties, all of which can be derived from the two properties above:
  - Identity element: $T(\mathbf{0}) = \mathbf{0}$
  - Superposition: $T\left(c_1\mathbf{u} + c_2\mathbf{v}\right) = c_1T(\mathbf{u}) + c_2T(\mathbf{v})$
- Common linear transformations include:
  - The **zero transformation** $T(\mathbf{v}) = \mathbf{0}$ for all $\mathbf{v}$ in the domain of T.
  - The **identity transformation** $T(\mathbf{v}) = \mathbf{v}$ for all $\mathbf{v}$ in the domain of T.
  - The **contraction** or **dilation** transformation $T(\mathbf{v}) = c\mathbf{v}$.

Introduction to Matrix Transformations

- The transformation $T(\mathbf{v}) = A\mathbf{v}$ is called a **matrix transformation**, where A is called the **standard matrix** of T.
  - An $m \times n$ standard matrix results in a linear transformation from $\mathbb{R}^n \to \mathbb{R}^m$.
    - In fact, *any* linear transformation from $\mathbb{R}^n \to \mathbb{R}^m$ can be represented as a matrix transformation with a UNIQUE $m \times n$ standard matrix:

$$T \text{ is linear} \iff \exists! A \text{ such that } T(\mathbf{v}) = A\mathbf{v} \text{ for all } \mathbf{v} \in \mathbb{R}^n.$$

  - The range of a matrix transformation $T(\mathbf{v}) = A\mathbf{v}$ is equal to the column space of A.
    The column space of A is the span of the columns of A.
- The columns of the standard matrix are what transformation does to the **standard basis vectors**.
  - The standard basis vectors in $\mathbb{R}^n$ are the vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ where $\mathbf{e}_i$ has a 1 in the i^{th} entry and 0 in all other entries. For example, in $\mathbb{R}^2$, the standard basis vectors are $\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.
    The standard basis vectors are the columns of the identity matrix I_n.
  - Multiplying a matrix A by the standard basis vector $\mathbf{e}_i$ gives us the i^{th} column of A: $A\mathbf{e}_i = \mathbf{a}_i$ where $\mathbf{a}_i$ is the i^{th} column of A.
  - If A is the standard matrix of a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$, then:

$$A = \left(T\left(\mathbf{e}_1\right) \;\; T\left(\mathbf{e}_2\right) \;\; \dots \;\; T\left(\mathbf{e}_n\right)\right).$$

    In other words, the i^{th} column of A is $T\left(\mathbf{e}_i\right)$.

One-to-One and Onto Transformations

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation with standard matrix A, so $T(\mathbf{v}) = A\mathbf{v}$.

- T is **one-to-one** or **injective** if every vector in the codomain is the image of *at most* one vector in the domain.
  - This is a question of *uniqueness*. If $\mathbf{x} = \mathbf{v}$ is a solution to $A\mathbf{x} = \mathbf{b}$, is $\mathbf{v}$ the only solution? Either it is the only solution or it has infinitely many solutions. The latter is only possible if there is a free variable, i.e. if there is a nonpivot column. Applying the contrapositive, **if all columns of A are pivot columns, then T is one-to-one**.
- T is **onto** or **surjective** if every vector in the codomain is the image of *at least* one vector in the domain. In other words, the range of T is equal to the codomain of T.
  - This is a question of *existence*. For every $\mathbf{b}$ in the codomain, does there exist a solution to $A\mathbf{x} = \mathbf{b}$? In other words, is the matrix $(A \;\; \mathbf{b})$ consistent for all $\mathbf{b}$? We cannot have a pivot in $\mathbf{b}$, so we *must* have a pivot in every row of A. So, **if all rows of A are pivot rows, then T is onto**.
- T is **bijective** if it is both **one-to-one and onto**, i.e. every vector in the codomain is the image of *exactly one* vector in the domain.
  - This is a question of *existence and uniqueness*. **If there is a pivot in every row and every column of A, then T is one-to-one and onto**.
  - If $n = m$, i.e. if $T : \mathbb{R}^n \to \mathbb{R}^n \iff A$ is a square matrix, then T is either *both* one-to-one and onto, or *neither* one-to-one nor onto.

Common Matrix Transformations

Recall that ALL linear transformations can be represented as matrix transformations.

- The **identity transformation** $\mathrm{Id}(\mathbf{v}) = \mathbf{v}$. This means $\mathrm{Id}(\mathbf{e}_1) = \mathbf{e}_1$, $\mathrm{Id}(\mathbf{e}_2) = \mathbf{e}_2$, …, $\mathrm{Id}(\mathbf{e}_n) = \mathbf{e}_n$.
  - The standard matrix of the identity transformation is the **identity matrix** I_n. It is the $n \times n$ square matrix with ones in the diagonal and zeros everywhere else.

$$\mathrm{Id}(\mathbf{v}) = I_n\mathbf{v} \quad \text{where} \quad I_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}_{n\times n}$$

- The **rotation transformation** $T(\mathbf{v}) = R\mathbf{v}$, where R is a rotation matrix. For example, in $\mathbb{R}^2$, the rotation transformation that rotates vectors by an angle of θ counterclockwise about the origin has the standard matrix:

$$R = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

- The **shear transformation** $T(\mathbf{v}) = S\mathbf{v}$, where:

$$S = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} \text{ or } S = \begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}.$$

One axis is fixed, while the other axis is shifted.

Summary

- Every linear transformation can be represented as a matrix transformation $T(\mathbf{v}) = A\mathbf{v}$, where A is the standard matrix of T.
  - To find A, apply T to the standard basis vectors $\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n$ and use the resulting vectors as the columns of A.
- The range of a matrix transformation is equal to the column space of its standard matrix.
- A matrix transformation with standard matrix A is:
  - **one-to-one** if A has a pivot in every column (satisfies uniqueness)
  - **onto** if A has a pivot in every row (satisfies existence)
  - **bijective** if A is both one-to-one and onto (satisfies uniqueness and existence)

2 Matrix Algebra

2.1 Matrix Operations

Matrix Multiplication

Let A be an $m \times p$ matrix and B be an $p \times n$ matrix. The **matrix product** AB is the $m \times n$ matrix where the entry in the i^{th} row and j^{th} column is given by:

$$(AB)_{ij} = \sum_{k=1}^{p} a_{ik} b_{kj}.$$

AB represents the transformation that results from first applying the transformation represented by B, followed by the transformation represented by A.

- In other words, to compute the entry in the i^{th} row and j^{th} column of the product matrix AB, we take the dot product of the i^{th} row of A with the j^{th} column of B:

$$(AB)_{ij} = \mathbf{a}_{i*} \cdot \mathbf{b}_{*j}.$$

- Another way to think about matrix multiplication is that each column of the product matrix AB is obtained by multiplying A by the corresponding column of B:

$$AB = \begin{pmatrix} A\mathbf{b}_1 & A\mathbf{b}_2 & \dots & A\mathbf{b}_n \end{pmatrix}$$

  where $\mathbf{b}_j$ is the j^{th} column of B for $j = 1, 2, \dots, n$.

- If A and B are matrices with appropriate dimensions, AB is the standard matrix of the transformation $T(U(\mathbf{v}))$ where $T(\mathbf{v}) = A\mathbf{v}$ and $U(\mathbf{v}) = B\mathbf{v}$.
  - The transformation with standard matrix A^n is the transformation that results from applying the transformation with standard matrix A n times in a row. This is only valid if A is square.

- Matrix multiplication does **not** commute. In other words, AB is not necessarily equal to BA, and in some cases, BA may not even be defined.

- Matrix multiplication is associative, so $(AB)C = A(BC)$.

- If $AB = AC$, $B = C$ **only if** A is invertible.
  Otherwise, $AB = AC$ does not necessarily imply $B = C$.

Algebraic Properties of Matrix Spaces

1. Commutative under addition: $A + B = B + A$.
2. Associative under addition: $(A + B) + C = A + (B + C)$.
3. Additive identity: There exists a zero matrix O such that $A + O = A$ for all matrices A.
4. Additive inverses: For every matrix A, there exists a matrix $-A$ such that $A + (-A) = O$.
5. Distributive under scalar multiplication: $c(A + B) = cA + cB$ and $\left(c_1 + c_2\right)A = c_1A + c_2A$.
6. Distributive under matrix multiplication: $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$.
7. Compatibility with scalar multiplication: $c_1\left(c_2A\right) = \left(c_1c_2\right)A$.

Properties of Matrix Transposes

Matrix Transpose

The **transpose** of an $m \times n$ matrix A, denoted $A^\top$, is the $n \times m$ matrix obtained by interchanging the rows and columns of A. In other words, the entry in the i^{th} row and j^{th} column of $A^\top$ is equal to the entry in the j^{th} row and i^{th} column of A:

$$\left(A^\top\right)_{ij} = A_{ji}.$$

- The transpose of an $m \times n$ matrix is an $n \times m$ matrix.
- It holds that $\left(A^\top\right)^\top = A$.
- If A and B have the same size, $(A + B)^\top = A^\top + B^\top$.
- If A is an $m \times p$ matrix and B is a $p \times n$ matrix, then $(AB)^\top = B^\top A^\top$ is an $n \times m$ matrix.

Properties of Inner and Outer Products

Inner Product of Two Vectors

The **inner product** of two (column) vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ is given by:

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^\top \mathbf{v} = u_1v_1 + u_2v_2 + \ldots + u_nv_n.$$

In other words, the inner product is simply the dot product.

Outer Product of Two Vectors

The **outer product** of two (column) vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ is given by:

$$\mathbf{u} \otimes \mathbf{v} = \mathbf{u}\mathbf{v}^\top = \begin{pmatrix} u_1v_1 & u_1v_2 & \ldots & u_1v_n \\ u_2v_1 & u_2v_2 & \ldots & u_2v_n \\ \vdots & \vdots & \ddots & \vdots \\ u_nv_1 & u_nv_2 & \ldots & u_nv_n \end{pmatrix}.$$

- Inner and outer products only exist for vectors with the same number of components.
- Only inner products commute: $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$.
- Outer products don't commute, flipping the order gives you the transpose: $\mathbf{u} \otimes \mathbf{v} = (\mathbf{v} \otimes \mathbf{u})^\mathsf{T}$.

Partitioned Matrices

2.2 The Inverse of a Matrix

- An $n \times n$ matrix A is **invertible** (or **nonsingular**) if there exists an $n \times n$ matrix B such that:

$$AB = I_n \text{ and } BA = I_n,$$

  where I_n is the $n \times n$ identity matrix.
  - The matrix B is called the **inverse** of A, denoted A^{-1}.
  - The inverse of a matrix is unique (if it exists).

- If a matrix is not invertible, it is called **singular**.

- If A is invertible, then the transformation $\mathbf{x} \mapsto A\mathbf{x}$ is both *one to one and onto*.
  - In other words, $\forall \mathbf{b} \in \mathbb{R}^n$, the equation $A\mathbf{x} = \mathbf{b}$ has *exactly one* solution given by $\mathbf{x} = A^{-1}\mathbf{b}$.

Proof: If A is invertible, then $A\mathbf{x} = \mathbf{b}$ has exactly one solution for $\mathbf{x}$ $\forall \mathbf{b}$

Let A be an invertible matrix with inverse A^{-1}. Then, for any $\mathbf{b}$ in $\mathbb{R}^n$, we can multiply both sides of the equation $A\mathbf{x} = \mathbf{b}$ by A^{-1} to get:

$$\begin{aligned} A^{-1}A\mathbf{x} &= A^{-1}\mathbf{b} \\ I_n\mathbf{x} &= A^{-1}\mathbf{b} \\ \mathbf{x} &= A^{-1}\mathbf{b}. \end{aligned}$$

Thus, there is at least one solution to the equation. To show that there is at most one solution, say $\mathbf{x}_1$ and $\mathbf{x}_2$ are two solutions to the equation. Then:

$$\begin{aligned} A\mathbf{x}_1 &= A\mathbf{x}_2 \\ A(\mathbf{x}_1 - \mathbf{x}_2) &= \mathbf{0}. \end{aligned}$$

Since A is invertible, the only solution to the equation $A\mathbf{x} = \mathbf{0}$ is the trivial solution $\mathbf{x} = \mathbf{0}$, so we must have $\mathbf{x}_1 - \mathbf{x}_2 = \mathbf{0}$, which means $\mathbf{x}_1 = \mathbf{x}_2$. Thus, there is at most one solution to the equation.

- An $m \times n$ matrix is **left invertible** if there exists an $n \times m$ matrix B such that $BA = I_n$. An $m \times n$ matrix is **right invertible** if there exists an $n \times m$ matrix C such that $AC = I_m$.
  - While *only* square matrices can be invertible, non-square matrices can be either left or right invertible, but not both.

Properties of Inverses

- Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be a 2×2 matrix. A is invertible iff $ad - bc \neq 0$, where its inverse is given by:

$$A^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

- If A is invertible, then A^{-1} is also invertible and its inverse is A: $\left(A^{-1}\right)^{-1} = A$.

- The inverse of a matrix transformation $T(\mathbf{x}) = A\mathbf{x}$ is $T^{-1}(\mathbf{x}) = A^{-1}\mathbf{x}$.
  - By the definition of an inverse, $T\left(T^{-1}(\mathbf{x})\right) = T^{-1}(T(\mathbf{x})) = \mathbf{x}$.

- $(AB)^{-1} = B^{-1}A^{-1}$ (the combined transformation happens in reverse order)
- $\left(A^{\top}\right)^{-1} = \left(A^{-1}\right)^{\top}$

Invertible Matrix Theorem

Let A be an $n \times n$ matrix. The following statements are equivalent (i.e. all true or all false):

1. A and/or A^{T} is invertible.
2. A is row-reducible to I_n.
3. A has n pivot positions.
4. The equation $A\mathbf{x} = \mathbf{0}$ only has the solution $\mathbf{x} = \mathbf{0}$.
5. The columns of A are linearly independent.
6. The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one and/or onto (in/sur/bijective).
7. The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b}$ in $\mathbb{R}^n$.
8. The columns of A span $\mathbb{R}^n$.
9. There exists an $n \times n$ matrix C such that $CA = I_n$.
10. There exists an $n \times n$ matrix D such that $AD = I_n$.

Elementary Matrices & Finding Inverses By Hand

- An **elementary matrix** that is at most one elementary row operation away from the associated identity matrix. All elementary matrices are invertible.
- Let E be the elementary matrix obtained by performing the elementary row operation R on the identity matrix. Then for any matrix A, EA is the matrix obtained by performing the elementary row operation R on A.
  - In other words, the elementary matrix is the standard matrix for the transformation that performs the row operation R.
  - Thus, it can be interpreted that an elementary matrix represents or corresponds with a certain elementay row operation.
- If a matrix A is invertible, then A is **row equivalent** to the identity matrix I.
  - In other words, there exists a sequence of elementary row operations that transforms A into I. When the same sequence of elementary row operations is applied to I, we get A^{-1}.

Proof: A sequence of row operations S s.t. $S(A) = I$ also satisfies $S(I) = A^{-1}$

Suppose A is invertible, so there exists a sequence of elementary row operations S such that $S(A) = I$. Let $E_1, E_2, \ldots, E_k$ be the elementary matrices corresponding to the row operations in S, so $S(A) = E_k \ldots E_2 E_1 A = I$. Then, we can multiply both sides of the equation by A^{-1} to get:

$$E_k \ldots E_2 E_1 A A^{-1} = I A^{-1}$$
$$E_k \ldots E_2 E_1 I = A^{-1}.$$

- To find the inverse of a matrix A, perform and record a sequence of elementary row operations to bring A into I. Then, apply the same sequence of elementary row operations to I to get A^{-1}.
  - To do these steps at the same time, we can invert matrix A by **row reducing the augmented matrix $\begin{pmatrix} A & I \end{pmatrix}$ to $\begin{pmatrix} I & A^{-1} \end{pmatrix}$**.

2.3 LU Factorization

- A matrix A can be factored into the product of a lower triangular matrix L and an upper triangular matrix U, such that $A = LU$. This is called the ***LU* factorization** of A.
  - The LU factorization of a matrix A can be used to solve the equation $A\mathbf{x} = \mathbf{b}$ by first solving $L\mathbf{y} = \mathbf{b}$ for $\mathbf{y}$ using forward substitution, and then solving $U\mathbf{x} = \mathbf{y}$ for $\mathbf{x}$ using back substitution.
    If we let $A = LU$ and $\mathbf{y} = U\mathbf{x}$, then $A\mathbf{x} = (LU)\mathbf{x} = L(U\mathbf{x}) = L\mathbf{y}$. So, the equation $A\mathbf{x} = \mathbf{b}$ becomes $L\mathbf{y} = \mathbf{b}$.

Lower Triangular Matrix L

Used for **forward** substitution, $L\mathbf{y} = \mathbf{b}$

$$\begin{pmatrix} * & 0 & 0 & \dots & 0 \\ * & * & 0 & \dots & 0 \\ * & * & * & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ * & * & * & \dots & * \end{pmatrix}$$

Upper Triangular Matrix U

Used for **back** substitution, $U\mathbf{x} = \mathbf{y}$

$$\begin{pmatrix} * & * & * & \dots & * \\ 0 & * & * & \dots & * \\ 0 & 0 & * & \dots & * \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & * \end{pmatrix}$$

The upper triangular matrix U can be obtained by reducing matrix A to row echelon form. Let the i^{th} row operation used to achieve U be represeted using the elementary matrix E_i. Then:

$$U = E_k \dots E_2 E_1 A.$$

If we let $A = LU$, then $U = E_k \dots E_2 E_1 A = L^{-1} A$.

Finding the LU Factorization of a Matrix

Any invertible matrix A can be factored into LU by reducing A to row echelon form to get U, and then using the row operations used to get U to construct L.

Start with the identity matrix I and the original matrix A. Reduce A to row echelon form U by performing **only row replacement operations**. Every time the operation $R_i \leftarrow R_i - kR_j$ is performed, record k as the entry in the i^{th} row of L, in the same column we are reducing.

A shortcut is to take the subcolumn of the column we are reducing, starting from the pivot and going down, and scale it by $\frac{1}{\text{pivot}}$ to get the entries in the corresponding column of L. This should be done *before* we begin reducing such column.

Example: Finding and using the LU factorization of a matrix

Find the LU factorization of the matrix $A = \begin{pmatrix} 2 & -1 & 1 \\ 3 & 3 & 9 \\ 4 & 2 & 2 \end{pmatrix}$, and use it to solve $A\mathbf{x} = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}$ for $\mathbf{x}$.

To find the LU factorization of A, we first reduce A to row echelon form to get U.

To reduce the first column:

2.4 Homogeneous Coordinates

- A transformation which shifts/translates points geometrically cannot be done with a matrix transformation. That is, a translation is not a linear transformation.
  - This is easy to see: $T(\mathbf{0}) = \mathbf{0}$ if T is linear. The origin cannot shift.
- **Homogeneous coordinates** add an extra dimension to ordinary coordinates so that they can capture translations in linear transformations.
  - The coordinate (x, y) in $\mathbb{R}^2$ is equivalent to the homogeneous coordinate $(x, y, 1)$ in $\mathbb{R}^3$
  - The homogeneous coordinate (x, y, z) is equivalent to the coordinate $\left(\frac{x}{z}, \frac{y}{z}\right)$ in $\mathbb{R}^2$ $(z \neq 0)$.

Homogeneous Coordinates

The coordinate $\left(x_1, x_2, \ldots, x_n\right)$ in $\mathbb{R}^n$ is equivalent to the homogeneous coordinate in $\mathbb{R}^{n+1}$:

$$\left(\lambda x_1, \lambda x_2, \ldots, \lambda x_n, \lambda\right) \text{ for any nonzero } \lambda.$$

  - Homogeneous coordinates work because they treat ordinary coordinates as a leveled "plane" in a higher dimension such that any translations in the plane can be reached from the origin.
- In homogeneous coordinates, we can represent translations as matrix transformations. For example, the translation that shifts points by $\binom{h}{k}$ can be represented by the matrix transformation with standard matrix:

$$\begin{pmatrix} 1 & 0 & h \\ 0 & 1 & k \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} x+h \\ y+k \\ 1 \end{pmatrix}.$$

Standard Translation Matrix

The **standard translation matrix** that shifts points in $\mathbb{R}^n$ by $\mathbf{h} = \left(h_1, h_2, \ldots, h_n\right)$ is given by the standard *homogeneous* transformation matrix:

$$A = \begin{pmatrix} I_n & \mathbf{h} \\ \mathbf{0} & 1 \end{pmatrix} \text{ where } I_n \text{ is the } n \times n \text{ identity matrix.}$$

It holds that $A\begin{pmatrix} \mathbf{x} \\ 1 \end{pmatrix} = \begin{pmatrix} \mathbf{x}+\mathbf{h} \\ 1 \end{pmatrix}$.

- For example, in $\mathbb{R}^3$, the translation that shifts points by (h, k, l) can be represented by the matrix transformation with standard matrix:

$$\begin{pmatrix} 1 & 0 & 0 & h \\ 0 & 1 & 0 & k \\ 0 & 0 & 1 & l \\ 0 & 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} x+h \\ y+k \\ z+l \\ 1 \end{pmatrix}.$$

- Any standard transformation matrix A has an equivalent matrix to transform homogeneous coordinates given by $\begin{pmatrix} A & \mathbf{0} \\ \mathbf{0} & 1 \end{pmatrix}$. For example, the standard rotation matrix $\boldsymbol{R}_\theta$ is equivalent to:

$$\begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Affine Transformations

Projective Transformations

- A **projection** or a **projective transformation** is a transformatio that maps points from $\mathbb{R}^n$ to a lower dimension $\mathbb{R}^m$, where $m < n$.
- A **perspective projection** is a type of projective transformation that maps points from $\mathbb{R}^n$ to $\mathbb{R}^{n-1}$. In $\mathbb{R}^3$, a perspective projection maps points from $\mathbb{R}^3$ to $\mathbb{R}^2$—that is—onto a plane.
  - A perspective projection in $\mathbb{R}^3$ maps a coordinate (x, y, z) onto an image point $\left(x^*, y^*, 0\right)$ such that a chosen **center point**, is colinear with the original point and its image.
    - In other words, if a projective projection of a point (x, y, z) through a center point (a, b, c) is $\left(x^*, y^*, 0\right)$, then we can draw line through $\left\{(x, y, z), (a, b, c), \left(x^*, y^*, 0\right)\right\}$.
  - This way, if the center point is (a, b, c), then the projection of the center point is just $(a, b, 0)$.
- Let a perspective projection in $\mathbb{R}^3$ have center point $(0, 0, d)$. Then, the perspective projection of a point (x, y, z) is given by the homogeneous matrix:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1/d & 1 \end{pmatrix}$$

3 Determinants

3.1 Introduction to Determinants

- The **determinant** of a square matrix is a scalar value which represents the factor by which the area/volume of a region is scaled when the linear transformation associated with the matrix is applied to that region.
- If the determinant of a matrix is zero, the linear transformation compresses the area/volume to zero and we cannot reverse the transformation (i.e., the matrix is not invertible).
  - In fact, the determinant is nonzero **iff** the matrix is invertible.

> **Determinant of a Matrix**
>
> The **determinant** ($\det A$ or $|A|$) of an $n \times n$ matrix A ($n \geq 2$) is:
>
> $$\det A = \sum_{j=1}^{n} (-1)^{1+j} a_{1j} \det\left(M_{1j}\right)$$
>
> where M_{1j} is the **minor** of A at row 1, column j, which is the $(n-1) \times (n-1)$ matrix that results from deleting row 1 and column j from A.

- The determinant of a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is $ad - bc$. (We can write $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$.)

Cofactor Expansion

We can actually expand along any row or column, not just the first row:

$$\det A = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} \det\left(M_{ij}\right) \qquad \text{expansion along } i^{\text{th}} \text{ row}$$

$$\det A = \sum_{i=1}^{n} (-1)^{i+j} a_{ij} \det\left(M_{ij}\right) \qquad \text{expansion along } j^{\text{th}} \text{ column}$$

All expansions will yield the same determinant. Notice the sign $(-1)^{i+j}$ follows a checkerboard pattern, allowing us to easily determine whether to add or subtract each term:

$$\begin{pmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{pmatrix}$$

The **cofactor** of an entry a_{ij} is defined as $c_{ij} = (-1)^{i+j} \det\left(M_{ij}\right)$, capturing what we multiply by a_{ij} in the determinant expansion. The **cofactor matrix** C of an $n \times n$ matrix A is the matrix where each entry c_{ij} is the cofactor of a_{ij}.

Computing determinants using cofactors is known as the method of **cofactor expansion**:

$$\det A = \sum_{i=1}^{n} a_{ij} c_{ij} \text{ for any } j = \sum_{j=1}^{n} a_{ij} c_{ij} \text{ for any } i.$$

Example: Using the Cofactor Expansion

Show that the cofactor expansion along the first row of a 3×3 matrix results in the same determinant as the expansion along the second column.

Let $A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$. The cofactor expansion along the first row is:

$$\begin{aligned} \det A &= a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix} \\ &= a(ei - fh) - b(di - fg) + c(dh - eg) \\ &= aei - afh - bdi + bfg + cdh - ceg. \end{aligned}$$

The cofactor expansion along the second column is:

$$\begin{aligned} \det A &= -b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + e \begin{vmatrix} a & c \\ g & i \end{vmatrix} - h \begin{vmatrix} a & c \\ d & f \end{vmatrix} \\ &= -b(di - fg) + e(ai - cg) - h(af - cd) \\ &= -bdi + bfg + eai - ecg - haf + hcd \\ &= aei - afh - bdi + bfg + cdh - ceg. \end{aligned}$$

Thus, both expansions yield the same determinant.

Note

Cofactor expansion is an inefficient method, as it requires computing the determinant of many smaller matrices. It is an $\mathcal{O}(n!)$ operation.

3.2 Properties of Determinants

Determinant of a Triangular Matrix

Let L be a lower triangular matrix, and let U be an upper triangular matrix:

$$L = \begin{pmatrix} l_1 & 0 & 0 & \dots & 0 \\ * & l_2 & 0 & \dots & 0 \\ * & * & l_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ * & * & * & \dots & l_n \end{pmatrix} \qquad U = \begin{pmatrix} u_1 & * & * & \dots & * \\ 0 & u_2 & * & \dots & * \\ 0 & 0 & u_3 & \dots & * \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & u_n \end{pmatrix}$$

Then:

$$\det L = l_1 l_2 l_3 \dots l_n.$$
$$\det U = u_1 u_2 u_3 \dots u_n.$$

That is, the determinant of a triangular matrix is the product of the entries on its main diagonal.

Proof: $\det L$ is the product of the entries on the main diagonal of L

Let L be a lower triangular matrix as defined above. We can compute $\det L$ by expanding along the first row:

$$\det L = l_1 \det\left(M_{11}\right) + 0 + 0 + \dots + 0 = l_1 \det\left(M_{11}\right).$$

The minor M_{11} is also a lower triangular matrix with main diagonal entries $l_2, l_3, \dots, l_n$. Thus, we can apply the same reasoning to compute $\det\left(M_{11}\right)$ by expanding along its first row, and so on until we reach a 1×1 matrix. This gives us:

$$\det L = l_1 \det\left(M_{11}\right) = l_1 l_2 \det\left(M_{22}\right) = \dots = l_1 l_2 \dots l_n.$$

Effects of Row Operations on the Determinant

- The three elementary row operations affect the determinant as follows:
  1. Swapping two rows multiplies the determinant by -1.
  2. Multiplying a row by a scalar k multiplies the determinant by k.
  3. Adding a multiple of one row to another row does not change the determinant.
- Using these properties, the determinant of a matrix is nonzero **iff** the matrix is row-reducible to a triangular matrix with no zeros on the main diagonal.
- $\det(kA) = k^n \det(A)$ for an $n \times n$ matrix A, as it is equivalent to multiplying each row of A by k.

Determinant of Matrix Products

Let A and B be $n \times n$ matrices. Then:

$$\det(AB) = \det A \det B.$$

Using these properties, the determinant of a matrix A can be computed by factorizing the matrix into $A = LU$, and then using $\det A = \det L \det U$ to compute the determinant of A in $\mathcal{O}\left(n^3\right)$ time, which is more efficient than the $\mathcal{O}(n!)$ time required by cofactor expansion.

Determinant of Matrix Transposes

- Transposing a matrix does not change its determinant:

$$\det A = \det A^{\mathsf{T}}$$

Determinant of Matrix Inverses

If A is invertible, then:

$$\det\left(A^{-1}\right) = \frac{1}{\det A}$$

Thus, if $\det A = 0$, then $\det\left(A^{-1}\right)$ is undefined and A^{-1} cannot exist. In fact, the determinant of a matrix is nonzero **iff** the matrix is invertible.

Proof: $\det\left(A^{-1}\right) = \frac{1}{\det A}$

Assume A is an invertible matrix. Then, we can write A as a product of elementary matrices acting on the identity I_n:

$$A = E_k \dots E_2 E_1 I_n \qquad\qquad A^{-1} = E_1^{-1} E_2^{-1} \dots E_k^{-1} I_n$$

where E_i are elementary matrices. Then:

$$\begin{aligned}\det\left(A^{-1}\right) &= \det\left(E_1^{-1} E_2^{-1} \dots E_k^{-1}\right) \\ &= \det\left(E_1^{-1}\right) \det\left(E_2^{-1}\right) \dots \det\left(E_k^{-1}\right).\end{aligned}$$

Each elementary matrix is invertible, and the determinant of an elementary matrix is either -1, k, or 1 depending on the type of row operation it corresponds to. Thus, the determinant of the inverse of an elementary matrix is either -1, $\frac{1}{k}$, or 1. Therefore, we have:

$$\begin{aligned}\det\left(A^{-1}\right) &= \det\left(E_1^{-1}\right) \det\left(E_2^{-1}\right) \dots \det\left(E_k^{-1}\right) \\ &= \left(\det E_1\right)^{-1} \left(\det E_2\right)^{-1} \dots \left(\det E_k\right)^{-1} \\ &= \frac{1}{\det E_1 \det E_2 \dots \det E_k} \\ &= \frac{1}{\det A}.\end{aligned}$$

Linearity of (a) Determinant Function

Let A be an $n \times n$ matrix, and let f be a function of each of the n column vectors of A, such that the domain of f is $\mathbb{R}^n$. Then, if we define f to be the following:

$$f(\mathbf{x}) = \det\left(\mathbf{a}_1 \ \ \mathbf{a}_2 \ \ \dots \ \ \mathbf{a}_{i-1} \ \ \mathbf{x} \ \ \mathbf{a}_{i+1} \ \ \dots \ \ \mathbf{a}_n\right)$$

Then $f : \mathbb{R}^n \to \mathbb{R}$ is a linear transformation. That is, we take the *determinant* of the matrix formed by replacing a particular chosen column, say the i^{th} column, of A with the input vector $\mathbf{x}$.

In other words, we can treat the determinant as a linear function **iff** all but one columns are fixed in the matrices we provide as inputs, in which case it is linear with respect to the unfixed column.

3.3 Cramer's Rule

Cramer's Rule

Let A be an invertible $n \times n$ matrix, and $\mathbf{b} \in \mathbb{R}^n$. Then the unique solution $\mathbf{x}$ to the equation $A\mathbf{x} = \mathbf{b}$ has entries given by:

$$x_i = \frac{\det A_i}{\det A} \text{ for } i = 1, 2, \ldots, n$$

where A_i is the matrix that results from replacing the i^{th} column of A with the column vector $\mathbf{b}$, or in other words:

$$\left(A_i\right)_{jk} = \begin{cases} b_j & \text{if } k = i \\ a_{jk} & \text{otherwise} \end{cases}$$

- To put it simply, Cramer's Rule allows us to solve for a particular variable in a system of linear equations.
  1. Take the determinant of the coefficient matrix A, this is the denominator ($\det A$).
  2. Replace the i^{th} column of A with the constants from the right-hand side of the equations to form A_i, and take its determinant, this is the numerator ($\det A_i$).
  3. The solution for the variable whose coefficients were in the i^{th} column (x_i) is $x_i = \frac{\det A_i}{\det A}$.
- Realize that the coefficient matrix must be square, i.e. we must have the same number of equations as unknowns.
  - Additionally, the coefficient matrix must be invertible (i.e., $\det A \neq 0$) for Cramer's Rule to be applicable. Otherwise, the system may have no solutions or infinitely many solutions.
- Cramer's Rule is most efficient for solving one variable in large systems, but may not be the best method for solving for all variables (since it involves calculating $n + 1$ determinants).

The Adjugate Matrix

Recall that if A is invertible, then the vector $\mathbf{x}_j$ whcih satisfies $A\mathbf{x}_j = \mathbf{e}_j$ is the j^{th} column of A^{-1}. This is because if $B = A^{-1}$, then $AB = I$ and $\left(A\mathbf{b}_1 \;\; A\mathbf{b}_2 \;\; \ldots \;\; A\mathbf{b}_n\right) = \left(\mathbf{e}_1 \;\; \mathbf{e}_2 \;\; \ldots \;\; \mathbf{e}_n\right)$.

By Cramer's Rule, the j^{th} column of A^{-1} is given by:

$$\left(A^{-1}\right)_{ij} = \frac{\det A_i}{\det A} \text{ for } i = 1, 2, \ldots, n$$

The j^{th} column of A_i is just $\mathbf{e}_j$, so a cofactor expansion down that column is:

$$\det A_i = 0 + 0 + \ldots + (-1)^{j+i}(1)\det\left(M_{ji}\right) + \ldots + 0 = (-1)^{j+i}\det\left(M_{ji}\right) = c_{ji} = \text{cofactor of } a_{ji}.$$

Note that it is the j^{th} row of $\mathbf{e}_j$ that is 1, and it is the i^{th} column that is replaced. Thus, if we transpose the cofactor matrix C, we get thee **adjugate** of a matrix A, denoted adj A. The entries of adj A are:

$$(\text{adj } A)_{ji} = c_{ij} = (-1)^{i+j}\det\left(M_{ij}\right) \text{ for } i, j = 1, 2, \ldots, n.$$

Finally, by dividing each cofactor in the adjugate by $\det A$, we get the entries of A^{-1}:

Inverse of a Matrix via Adjugates

If A is an invertible $n \times n$ matrix and $\operatorname{adj} A$ is its **adjugate**, then:

$$A^{-1} = \frac{\operatorname{adj} A}{\det A}.$$

4 Vector Spaces

4.1 Introduction to Subspaces

- A **vector space** V is a nonempty set of objects called *vectors* defined under the following axioms:

Axioms of Vector Spaces

*Define two closed operations on V: **vector addition** and **scalar multiplication**.*

1. $\forall \mathbf{u}, \mathbf{v} \in V, \mathbf{u} + \mathbf{v} \in V$ (closed under addition)
2. $\forall \mathbf{u}, \mathbf{v} \in V, \mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ (vector addition is commutative)
3. $\forall \mathbf{u}, \mathbf{v}, \mathbf{w} \in V, (\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ (vector addition is associative)
4. $\exists \mathbf{0} \in V$ s.t. $\forall \mathbf{v} \in V, \mathbf{v} + \mathbf{0} = \mathbf{0} + \mathbf{v} = \mathbf{v}$ (existence of additive identity)
5. $\forall \mathbf{v} \in V, \exists (-\mathbf{v}) \in V$ s.t. $\mathbf{v} + (-\mathbf{v}) = (-\mathbf{v}) + \mathbf{v} = \mathbf{0}$ (existence of additive inverses)
6. $\forall k \in \mathbb{R}, \forall \mathbf{v} \in V, k\mathbf{v} \in V$ (closed under scalar multiplication)
7. $\forall k \in \mathbb{R}, \forall \mathbf{u}, \mathbf{v} \in V, k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$ (distributive property 1)
8. $\forall m, n \in \mathbb{R}, \forall \mathbf{v} \in V, (m + n)\mathbf{v} = m\mathbf{v} + n\mathbf{v}$ (distributive property 2)
9. $\forall m, n \in \mathbb{R}, \forall \mathbf{v} \in V, (mn)\mathbf{v} = m(n\mathbf{v})$ (scalar multiplication is associative)
10. $\forall \mathbf{v} \in V, 1\mathbf{v} = \mathbf{v}$ (scalar multiplicative identity)

- A **subspace** H of a vector space V is a subset of V that is itself a vector space.
  - H must satisfy the ten axioms of vector spaces under the *same* vector addition and scalar multiplication defined on V:
    - Vector addition and scalar multiplication must *also* be closed on H.
    - The zero vector must be *also* an element of H.

Subspace

A **subspace** H of a vector space V is a subset of V such that H is itself a vector space under the same vector addition and scalar multiplication defined on V.

If $H \subseteq V$, then H is a subspace of V **iff** H follows three conditions:
1. The zero vector of V is in H.
2. H is closed under vector addition: $\forall \mathbf{u}, \mathbf{v} \in H, \mathbf{u} + \mathbf{v} \in H$.
3. H is closed under scalar multiplication: $\forall k \in \mathbb{R}, \forall \mathbf{v} \in H, k\mathbf{v} \in H$.

- $\{\mathbf{0}\}$ is a subspace of every vector space, and every vector space is a subspace of itself.
- The span of any set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$ in a vector space V is a subspace of V. Conversely, every subspace of V can be expressed as the span of a set of p vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$ from V.
  - We say V is the subspace *spanned by* or *generated by* $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$.
  - We call $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$ the *spanning set* of V.

Special Subspaces

Column Space

The **column space** of an $m \times n$ matrix A, denoted $\mathrm{col}(A)$, is the set of all linear combinations of the columns of A, i.e. the span of the columns of A. **It holds that col(A) is a subspace of $\mathbb{R}^m$.**

- If $\mathrm{col}(A) = \mathbb{R}^m$, then the transformation $\mathbf{v} \mapsto A\mathbf{v}$ is surjective (onto).

Null Space and Kernel

The **null space** of an $m \times n$ matrix A, denoted $\mathrm{nul}(A)$, is the set of all solutions to the homogeneous equation $A\mathbf{x} = \mathbf{0}$. **It holds that nul(A) is a subspace of $\mathbb{R}^n$.**

The **kernel** of a *linear transformation* $T : V \to W$ is the set of all vectors in its domain V that are mapped to the zero vector in its codomain W:

$$\ker(T) = \{\mathbf{v} \in V \mid T(\mathbf{v}) = \mathbf{0}\}.$$

- $\ker(T)$ is a subspace of its domain V.
- If A is the standard matrix of T, then $\ker(T) = \mathrm{nul}(A)$.
- If $\ker(T) = \{\mathbf{0}\}$, then T is injective (one-to-one).

4.2 Basis Vectors, Rank, & Dimension

Basis

A **basis** $\mathcal{B}$ of a vector space V is a linearly independent subset of V such that $\mathcal{B}$ spans V:

$$\text{span}(\mathcal{B}) = V \text{ where } \mathcal{B} \subseteq V.$$

That is, we can take any linear combination of vectors in $\mathcal{B}$ to get any vector in V, and no vector in $\mathcal{B}$ can be written as a linear combination of the other vectors in $\mathcal{B}$.

- A basis for the column space of a matrix A can be found by taking the columns of A that correspond to the pivot columns in the row echelon form of A. That is, *a basis for* $\text{col}(A)$ *are the pivot columns of* A.

  Note: *the basis is formed from the columns in the original A, not the reduced form of A!*

Dimension of a Vector Space

The **dimension** of a vector space V is the number of vectors in a basis for V. That is, it is the minimum number of vectors needed to span V. We denote the dimension of V as $\dim(V)$.

If $\dim(V) = n$, we call the vector space V *n-dimensional*.

- It holds that although a vector space may have many different bases, **all bases for a vector space have the same number of vectors**. In other words, *a vector space has a well-defined dimension*.
  This is referred to as the **Unique Size Theorem**.
  - If we can find *any* linearly independent set of n vectors S in a vector space V with dimension n, then S is a basis for V (i.e. we imply that S spans V)
  - If we can find *any* spanning set of n vectors S in a vector space V with dimension n, then S is a basis for V (i.e. we imply linear independence of S)

  Collectively, the above two properties are known as the **Basis Theorem**.
- For any subspace H of V:
  - It holds that $\dim(H) \leq \dim(V)$.
    - The smallest subspace is $\{\mathbf{0}\}$, which has dimension 0 because its basis is the empty set $\varnothing$.
  - For any basis for H, say $\mathcal{B}_H$, there exists a basis for V, say $\mathcal{B}_V$, which is a superset of $\mathcal{B}_H$.
    In other words, we can always find a basis for V by "extending" any basis for H:

    $$\mathcal{B}_H \subseteq \mathcal{B}_V \text{ where } \mathcal{B}_H \text{ is a basis for } H \text{ and } \mathcal{B}_V \text{ is a basis for } V.$$
- The dimension of $\mathbb{R}^n$ is simply n, as the standard unit vectors $\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n$ form a basis for $\mathbb{R}^n$.

Rank and Nullity of a Matrix

The **rank** of a matrix A, denoted $\mathrm{rank}(A)$, is the dimension of the column space of A (or equivalently, the row space of A). It is the number of pivot positions in A:

$$\mathrm{rank}(A) = \dim(\mathrm{col}(A)) = \dim(\mathrm{row}(A)) = \text{number of pivot positions in } A.$$

The **nullity** of a matrix A, denoted $\mathrm{nullity}(A)$, is the dimension of the null space of A. It is the number of non-pivot columns in A (which correspond to the number of free variables in the homogeneous equation $A\mathbf{x} = \mathbf{0}$):

$$\mathrm{nullity}(A) = \dim(\mathrm{nul}(A)) = \text{number of non-pivot columns in } A.$$

If A is an $m \times n$ matrix, then:

- $\mathrm{rank}(A) = \mathrm{rank}\left(A^\top\right)$.
- $\mathrm{rank}(A) \leq \min(m, n)$.
- $\mathrm{rank}(AB) \leq \min(\mathrm{rank}(A), \mathrm{rank}(B))$ for some $n \times p$ matrix B.
- $\mathrm{rank}(A) = n$ means:
  - the equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{x} = \mathbf{0}$, so $\mathrm{nul}(A) = \{\mathbf{0}\}$
  - the transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.
- $\mathrm{rank}(A) = m$ means:
  - the equation $A\mathbf{x} = \mathbf{b}$ has a solution for every $\mathbf{b} \in \mathbb{R}^m$.
  - the transformation $\mathbf{x} \mapsto A\mathbf{x}$ is onto.

Rank-Nullity Theorem

Let A be an $m \times n$ matrix. Then:

$$\mathrm{rank}(A) + \mathrm{nullity}(A) = n.$$

This is simply saying that "*# pivot columns in A + # non-pivot columns in A = # columns in A.*"

Invertible Matrix Theorem *(continued)*

Let A be an $n \times n$ matrix. Then the following statements are equivalent:

1. Everything from the The first part of the Invertible Matrix Theorem.
2. The columns of A form a basis for $\mathbb{R}^n$.
3. The rows of A form a basis for $\mathbb{R}^n$.
4. $\mathrm{col}(A) = \mathrm{row}(A) = \mathbb{R}^n$.
5. $\mathrm{rank}(A) = n$.
6. $\mathrm{nul}(A) = \{\mathbf{0}\}$.
7. $\mathrm{nullity}(A) = 0$.
8. $\det(A) \neq 0$. *(from Section 3.1)*
9. 0 is not an eigenvalue of A. *(from Section 5.1)*

4.3 Basis Coordinates & Change of Basis

Basis Coordinates

- The **standard basis** for $\mathbb{R}^n$ is an ordered basis $\mathcal{E} = (\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n)$.
  - We call $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ the **standard basis vectors** for $\mathbb{R}^n$.

Note

An **ordered basis** is a basis represented as an ordered tuple rather than a set, such that the order of coordinates relative to that basis is well-defined.

- Given an ordered basis $\mathcal{B} = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$ for a vector space V, *any* vector in the vector space can be written as a **unique** linear combination of the basis vectors. This the **Unique Representation Theorem**.
  - If $\mathbf{v} \in V$, then there exist *unique* scalars $c_1, c_2, \dots, c_n$ such that:

$$\mathbf{v} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + \dots + c_n\mathbf{b}_n.$$

  In which case we can call the scalars $c_1, c_2, \dots, c_n$ the **coordinates** of $\mathbf{v}$ relative to the basis $\mathcal{B}$. The **coordinate vector** of $\mathbf{v}$ relative to $\mathcal{B}$ is denoted $[\mathbf{v}]_{\mathcal{B}}$ and is given by:

$$[\mathbf{v}]_{\mathcal{B}} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}.$$

  - The linear transformation $\mathbf{v} \mapsto [\mathbf{v}]_{\mathcal{B}}$ is the **change of basis transformation** from the standard basis $\mathcal{E}$ to the basis $\mathcal{B}$. It holds that this transformation is **one-to-one and onto** (bijective). We could also say that the change of basis transformation is an **isomorphism** between V and $\mathbb{R}^n$.
    - This fact allows us to treat any vector space as if it were $\mathbb{R}^n$ by using the change of basis transformation to convert vectors in the vector space to their coordinate vectors in $\mathbb{R}^n$.

Example: Finding Coordinates Relative to a Basis

What is the coordinate vector of $\mathbf{v} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$ relative to the basis $\mathcal{B} = \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix}\right)$?

We can write $\mathbf{v}$ as a linear combination of the basis vectors:

$$\mathbf{v} = c_1\begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} c_1 + c_2 \\ c_2 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}.$$

Solving this system gives us $c_1 = 1$ and $c_2 = 2$. Thus:

$$[\mathbf{v}]_{\mathcal{B}} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Changing Between Bases

- Given two ordered bases $\mathcal{B} = \left(\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n\right)$ and $\mathcal{C} = \left(\mathbf{c}_1, \mathbf{c}_2, \ldots, \mathbf{c}_n\right)$ for a vector space V, the **change-of-basis matrix** from $\mathcal{B}$ to $\mathcal{C}$, usually denoted P, is the matrix whose columns are the coordinate vectors of the basis vectors of $\mathcal{B}$ relative to the basis $\mathcal{C}$:

$$P_{\mathcal{B}\to\mathcal{C}} = \left(\left[\mathbf{b}_1\right]_{\mathcal{C}} \;\; \left[\mathbf{b}_2\right]_{\mathcal{C}} \;\; \cdots \;\; \left[\mathbf{b}_n\right]_{\mathcal{C}}\right)$$

  For example, the change-of-basis matrix from $\mathcal{B}$ to the standard basis $\mathcal{E}$ is just the columns of $\mathcal{B}$:

$$P_{\mathcal{B}\to\mathcal{E}} = \left(\mathbf{b}_1 \;\; \mathbf{b}_2 \;\; \ldots \;\; \mathbf{b}_n\right).$$

- Given a $\mathcal{B}$-coordinate $[\mathbf{v}]_{\mathcal{B}}$, we can find the $\mathcal{C}$-coordinate $[\mathbf{v}]_{\mathcal{C}}$ by multiplying the change-of-basis matrix from $\mathcal{B}$ to $\mathcal{C}$ by the $\mathcal{B}$-coordinate:

$$[\mathbf{v}]_{\mathcal{C}} = P_{\mathcal{B}\to\mathcal{C}}[\mathbf{v}]_{\mathcal{B}}.$$

  Change-of-basis matrices are always invertible, and the inverse of the change-of-basis matrix from $\mathcal{B}$ to $\mathcal{C}$ is the change-of-basis matrix from $\mathcal{C}$ to $\mathcal{B}$:

$$P_{\mathcal{C}\to\mathcal{B}} = \left(P_{\mathcal{B}\to\mathcal{C}}\right)^{-1}.$$

Example: Changing Coordinates Between Bases

Let $\mathcal{B} = \left(\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}1\\1\end{pmatrix}\right)$ and $\mathcal{C} = \left(\begin{pmatrix}2\\1\end{pmatrix}, \begin{pmatrix}0\\k\end{pmatrix}\right)$ be two bases for $\mathbb{R}^2$, for some $k \in \mathbb{R}$.

a. What is the change-of-basis matrix from $\mathcal{B}$ to $\mathcal{C}$?
b. What is the change-of-basis matrix from $\mathcal{C}$ to $\mathcal{B}$?
c. If the $\mathcal{B}$-coordinate of a vector is $\begin{pmatrix}1\\2\end{pmatrix}$, what is its $\mathcal{C}$-coordinate?

---

a.

4.4 Difference Equations

Recall:

- The set of solutions to $A\mathbf{x} = \mathbf{0}$ form a vector space.
  The set of solutions to $A\mathbf{x} = \mathbf{b}$, where $\mathbf{b} \neq \mathbf{0}$, can *never* be vector space.
- If $\mathbf{x} = \mathbf{x}_1$ and $\mathbf{x} = \mathbf{x}_2$ are solutions to $A\mathbf{x} = \mathbf{b}$, then $\mathbf{x}_1 - \mathbf{x}_2$ is *also* a solution to $A\mathbf{x} = \mathbf{0}$.
  That is, $\mathbf{x}_1 - \mathbf{x}_2 \in \text{nul}(A)$.
- The set of all solutions to a non-homogeneous equation $A\mathbf{x} = \mathbf{b}$ can be found by finding one solution $\mathbf{x} = \mathbf{p}$ and then adding any solution to $A\mathbf{x}_0 = \mathbf{0}$:

$$\{\mathbf{x} \mid A\mathbf{x} = \mathbf{b}\} = \left\{\mathbf{p} + \mathbf{x}_0 \mid A\mathbf{x}_0 = \mathbf{0}\right\}.$$

A **linear recurrence** is a sequence where each term is a linear combination of the previous terms. For example, the sequence $(2, 4, 8, 16, \ldots)$ is a linear recurrence.

Take the linear recurrence given by the **difference equation**:

$$y_{k+1} = 2y_k \quad \text{or} \quad y_{k+1} - 2y_k = 0.$$

Let $\mathbb{S}$ be the set of linear recurrences of the form $y_{k+1} = 2y_k$. Define the transformation $T : \mathbb{S} \to \mathbb{S}$ by $y_k \mapsto y_{k+1} - 2y_k$, such that $\ker(T) = \left\{\mathbf{y} \in \mathbb{S} \mid y_{k+1} = 2y_k\right\}$. That is, the kernel of our transformation is the set of solutions to the difference equation.

Specifically, the kernel is the set of all linear recurrences where each term is exactly double the previous term, i.e. recurrences of the form $\mathbf{y} = \left\{y_k = a2^k\right\}$ (for a is some constant/“initial value”).

In fact, we can say that the kernel is *spanned by* the basis $\left\{2^k\right\}$, and that $\ker(T)$ is a 1-dimensional subspace of $\mathbb{S}$.

The set of solutions to an n^{th} order homogeneous linear recurrence form an n-dimensional vector space.

Solutions to a Homogeneous Linear Recurrence

To find the set of solutions to an n^{th} order linear recurrence of the form:

$$y_{k+n} + a_{n-1}y_{k+n-1} + a_{n-2}y_{k+n-2} + \ldots + a_0 y_k = 0,$$

Find the roots $\left\{r_1, r_2, \ldots, r_n\right\}$ of the **characteristic polynomial**:

$$r^n + a_{n-1}r^{n-1} + a_{n-2}r^{n-2} + \ldots + a_0.$$

Then, a basis for the set of solutions to the linear recurrence is:

$$\mathcal{B} = \left\{r_1^k, r_2^k, \ldots, r_n^k\right\}.$$

That is, if $\mathbf{y}$ is a solution to the linear recurrence, then $\mathbf{y} \in \text{span}(\mathcal{B})$.

- in general: find the set of solutions to homogeneous case, then find a constant solution for nonhomoengeous, then add together.

A set of linear recurrences is **linearly independent** if each set of corresponding elements in each recurrence is a linearly independent set. That is, we

The **Casorati matrix** is a matrix used to determine the linear independence of a set of linear recurrences. Given linear recurrences $\mathbf{a}, \mathbf{b}, \mathbf{c}$, the Casorati matrix starting at index k is defined as:

$$C_k = \begin{pmatrix} a_k & b_k & c_k \\ a_{k+1} & b_{k+1} & c_{k+1} \\ a_{k+2} & b_{k+2} & c_{k+2} \end{pmatrix}.$$

5 Eigeneverything

5.1 Eigenvectors and Eigenvalues

Eigenvector and Eigenvalue

Let A be an $n \times n$ matrix. A nonzero vector $\mathbf{v} \in \mathbb{R}^n$ is an **eigenvector** of A if there exists a scalar λ such that:

$$A\mathbf{v} = \lambda\mathbf{v}.$$

The scalar λ is called the **eigenvalue** corresponding to the eigenvector $\mathbf{v}$. A may have multiple pairs of eigenvectors and eigenvalues.

- The eigenvalue equation is a homogeneous matrix equation with respect to $A - \lambda I$. It *must* have free variables such that we don't get trivial solutions (which wouldn't count as eigenvectors).
- An eigenvalue will have infinitely many associated eigenvectors. The set of all eigenvectors corresponding to a particular eigenvalue λ, together with the zero vector, form a subspace called the **eigenspace** of A corresponding to λ.
- The set of eigenvectors from *distinct* eigenvalues are linearly independent. (proof)
  - That is, let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be distinct eigenvalues of a matrix A.

    Suppose $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are eigenvectors corresponding to $\lambda_1, \lambda_2, \ldots, \lambda_n$ respectively. Then, the set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is linearly independent.
- The eigenvalues of a triangular matrix are the entries on its main diagonal.
- Although *eigenvectors* cannot be $\mathbf{0}$, *eigenvalues* can be 0, but only if its matrix is not invertible. That is, 0 is an eigenvalue of A **iff** A is not invertible.

Proof: Linear Independence of Eigenvectors from distinct Eigenvalues

Suppose a matrix A has distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$.
Let $\mathbf{v}_i$ be an eigenvector for λ_i, so $A\mathbf{v}_i = \lambda_i \mathbf{v}_i$.

Assume for the sake of contradiction that the set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is linearly dependent. Then, there exists some $\mathbf{v}_j$ $(1 < j \leq n)$ such that $\mathbf{v}_j$ can be written as a linear combination of its preceding eigenvectors:

$$\mathbf{v}_j = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_{j-1}\mathbf{v}_{j-1}$$

Then:

$$\begin{aligned} A\mathbf{v}_j &= A\left(c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_{j-1}\mathbf{v}_{j-1}\right) && \text{by left-multiplying both sides by } A \\ A\mathbf{v}_j &= c_1A\mathbf{v}_1 + c_2A\mathbf{v}_2 + \dots + c_{j-1}A\mathbf{v}_{j-1} && \text{by distributing } A \text{ on the rhs} \\ \lambda_j\mathbf{v}_j &= c_1\lambda_1\mathbf{v}_1 + c_2\lambda_2\mathbf{v}_2 + \dots + c_{j-1}\lambda_{j-1}\mathbf{v}_{j-1} && \text{by performing the substitution } A\mathbf{v}_i = \lambda_i\mathbf{v}_i \end{aligned}$$

Also:

$$\begin{aligned} \lambda_j\mathbf{v}_j &= \lambda_j\left(c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_{j-1}\mathbf{v}_{j-1}\right) && \text{by multiplying both sides by } \lambda_j \\ \lambda_j\mathbf{v}_j &= c_1\lambda_j\mathbf{v}_1 + c_2\lambda_j\mathbf{v}_2 + \dots + c_{j-1}\lambda_j\mathbf{v}_{j-1} && \text{by distributing } \lambda_j \text{ on the rhs} \end{aligned}$$

Finally, equate the two expressions for $\lambda_j\mathbf{v}_j$:

$$c_1\lambda_1\mathbf{v}_1 + c_2\lambda_2\mathbf{v}_2 + \dots + c_{j-1}\lambda_{j-1}\mathbf{v}_{j-1} = c_1\lambda_j\mathbf{v}_1 + c_2\lambda_j\mathbf{v}_2 + \dots + c_{j-1}\lambda_j\mathbf{v}_{j-1}$$

$$c_1\left(\lambda_1 - \lambda_j\right)\mathbf{v}_1 + c_2\left(\lambda_2 - \lambda_j\right)\mathbf{v}_2 + \dots + c_{j-1}\left(\lambda_{j-1} - \lambda_j\right)\mathbf{v}_{j-1} = \mathbf{0}.$$

However, all λ_i are distinct, so $\lambda_a - \lambda_b \neq 0$ for $a \neq b$. Thus, the above equation can only be satisfied if $c_1 = c_2 = \dots = c_{j-1} = 0$, which contradicts our assumption that $\mathbf{v}_j$ is a linear combination of its preceding eigenvectors. Therefore, the set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ must be linearly independent. ∎

From this theorem, we can derive the following:

- If an $n \times n$ matrix A has n distinct eigenvalues, then a basis for $\mathbb{R}^n$ can be formed from the corresponding eigenvectors of those eigenvalues, so A is diagonalizable.

  In other words, *if an $n \times n$ matrix has n distinct eigenvalues, then it is diagonalizable*.

- If A has less than n distinct eigenvalues, then a basis for $\mathbb{R}^n$ can be formed from the corresponding eigenvectors of those eigenvalues **only if** the dimension of each corresponding eigenspace has a dimension equal to the multiplicity of that eigenvalue.

  Then, we can take enough linearly inependent eigenvectors from each eigenspace to form a basis for $\mathbb{R}^n$, so A is diagonalizable.

5.2 The Characteristic Equation

The **characteristic equation** of an $n \times n$ matrix A is the polynomial equation of degree n of the form:

$$\det(A - \lambda I) = 0.$$

The solutions to this equation $\lambda_1, \lambda_2, \dots$ (that is, the roots of $\det(A - \lambda I)$) are the eigenvalues of A.

Derivation: Characteristic Equation

Let A be an $n \times n$ matrix, and let $\mathbf{v}$ be an eigenvector of A with corresponding eigenvalue λ. Then:

1. By the definitions of eigenvector and eigenvalue, $A\mathbf{v} = \lambda\mathbf{v}$.
2. Rearranging gives $A\mathbf{v} - \lambda\mathbf{v} = \mathbf{0}$.
3. It can be shown that for a column vector $\mathbf{v}$ and scalar k, $k\mathbf{v} = kI\mathbf{v}$, where $I = I_n$ is the $n \times n$ identity matrix.
4. Thus, $A\mathbf{v} - \lambda I\mathbf{v} = \mathbf{0}$, and by the distributive property, we have $(A - \lambda I)\mathbf{v} = \mathbf{0}$.
5. Since $\mathbf{v}$ is nonzero, our homogeneous equation has a nontrivial solution, so by the Invertible Matrix Theorem, $A - \lambda I$ cannot be invertible. Thus, $\det(A - \lambda I) = 0$.

For any given eigenvalue λ_i, any vector $\mathbf{v}_i$ which satisfies the homogeneous equation $\left(A - \lambda_i I\right)\mathbf{v} = \mathbf{0}$ is an eigenvector for A corresponding to λ_i.

That is, the eigenspace of A corresponding to λ_i is $\operatorname{nul}\left(A - \lambda_i I\right)$.

Steps to find Eigenvalues and Eigenvectors

For a square matrix A:

1. Find the eigenvalues of A by solving the characteristic equation $\det(A - \lambda I) = 0$ for $\lambda_1, \lambda_2, \dots, \lambda_n$.
2. For each eigenvalue λ_i, find the corresponding eigenvectors by solving the homogeneous equation $\left(A - \lambda_i I\right)\mathbf{v} = \mathbf{0}$ for $\mathbf{v}$.

Trace of a Matrix

The **trace** of an $n \times n$ matrix A, denoted $\operatorname{tr}(A)$, is the sum of the entries on the main diagonal of A.

- The trace of a matrix is equal to the sum of its eigenvalues, counting multiplicities.
- The characteristic equation of a 2×2 matrix is:

$$\det(A - \lambda I) = \lambda^2 - \operatorname{tr}(A)\lambda + \det(A) = 0$$

5.3 Diagonalization

A diagonal matrix (a matrix with only entries on its main diagonal) is easy to work with. Particularly:

$$D = \begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix} \to D^k = \begin{pmatrix} \lambda_1^k & & & \\ & \lambda_2^k & & \\ & & \ddots & \\ & & & \lambda_n^k \end{pmatrix}.$$

Diagonalization Theorem

An $n \times n$ matrix A is diagonalizable **iff** A has n linearly independent eigenvectors.

In other words, A is diagonalizable **iff** there exists a basis for $\mathbb{R}^n$ consisting of eigenvectors of A.

Let P be the matrix whose columns are the n linearly independent eigenvectors of A:

$$P = \begin{pmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \dots & \mathbf{v}_n \end{pmatrix}.$$

Let D be the diagonal matrix whose entries on the main diagonal are the corresponding eigenvalues of those eigenvectors:

$$D = \begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix}.$$

Then $\mathbf{v}_i$ is the eigenvector corresponding to λ_i, so $A\mathbf{v}_i = \lambda_i \mathbf{v}_i$. Then:

$$AP = \begin{pmatrix} A\mathbf{v}_1 & A\mathbf{v}_2 & \dots & A\mathbf{v}_n \end{pmatrix} = \begin{pmatrix} \lambda_1\mathbf{v}_1 & \lambda_2\mathbf{v}_2 & \dots & \lambda_n\mathbf{v}_n \end{pmatrix} = PD.$$

Since the columns of P are linearly independent under the Diagonalization Theorem, P is invertible by the Invertible Matrix Theorem. So, we can right-multiply both sides by P^{-1} to get:

$$A = PDP^{-1}.$$

This is the **diagonalization** of A. Again, the columns of P are the eigenvectors of A, and the entries on the main diagonal of D are the corresponding eigenvalues.

Diagonalization of a Matrix

An $n \times n$ matrix A is **diagonalizable** if there exists an invertible matrix P and a diagonal matrix D such that:

$$A = PDP^{-1}$$

where, given λ_i is the eigenvalue for the i^{th} eigenvector $\mathbf{v}_i$ of A:

$$P = \begin{pmatrix} \vdots & \vdots & \cdots & \vdots \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ \vdots & \vdots & \cdots & \vdots \end{pmatrix} \qquad D = \begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix}.$$

The columns of P are the eigenvectors of A, and the entries on the main diagonal of D are the corresponding eigenvalues.

- Realize that because P needs to be invertible, the columns of P must be linearly independent, so A must have n linearly independent eigenvectors in order to be diagonalizable.

  Recall that from this proof that this can follow if A has n distinct eigenvalues.

- If A is diagonalizable, then we can find an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$. Then, for any positive integer k, we have:

  $$A^k = \left(PDP^{-1}\right)^k = PD^kP^{-1}.$$

  This is much easier to compute than A^k directly, since D^k is just the diagonal matrix with the entries on the main diagonal raised to the power of k.

Example: Diagonalizing a Matrix

For example, if A has eigenvalues $\lambda_1 = 3, \lambda_2 = 2, \lambda_3 = 5$ with corresponding eigenvectors $\mathbf{v}_1 = \langle 1, 0, 0\rangle, \mathbf{v}_2 = \langle 0, 1, 0\rangle, \mathbf{v}_3 = \langle 0, 0, 1\rangle$ respectively, then:

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad D = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

Repeated Eigenvalues

Multiplicity of an Eigenvalue

The **multiplicity** of an eigenvalue λ is the number of times λ appears as a root of the characteristic polynomial $\det(A - \lambda I)$.

If λ is an eigenvalue of A with multiplicity m, then the dimension of the eigenspace corresponding to λ is at least 1 and at most m. In other words, there are at least 1 and at most m linearly independent eigenvectors corresponding to λ.

Diagonalizability of a Matrix with Repeated Eigenvalues

Let A be an $n \times n$ matrix with deduplicated eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_k$ with multiplicities $m_1, m_2, \ldots, m_k$ respectively. Then, A is diagonalizable **iff** the dimension of the eigenspace corresponding to each eigenvalue λ_i is equal to its multiplicity m_i for all $i = 1, 2, \ldots, k$.

In other words, A is diagonalizable **iff** there are enough linearly independent eigenvectors corresponding to each eigenvalue to form a basis for $\mathbb{R}^n$. For the eigenspace corresponding to λ_i, take m_i linearly independent vectors from the eigenspace corresponding to λ_i.

Similar Matrices

- For two matrices A and D, if $\exists$ an invertible matrix P such that $A = PDP^{-1}$, then A is **similar** to D. Equivalently, D is similar to A, so $D = QAQ^{-1}$ for some matrix $Q = P^{-1}$.
- Matrix similarity follows an equivalence relationship (reflexive, symmetric, and transitive)
- Similar matrices have the same characteristic polynomial and thus the same eigenvalues.

5.4 Eigenvectors in Linear Transformations

5.5 Complex Eigenvalues

- Where as real eigenvalues geometrically describe vectors that *scale*, complex eigenvalues describe vectors that both *rotate* and *scale*.
- If A is a real matrix, then the characteristic polynomial $\det(A - \lambda I)$ has real coefficients, so the complex eigenvalues of A come in conjugate pairs. That is, if $a + bi$ is an eigenvalue of A, then its complex conjugate $a - bi$ is also an eigenvalue of A. In such case, their eigenvectors will also be complex conjugates.

Block Diagonalization

Although a matrix with complex eigenvalues cannot be diagonalizable (to a real-valued matrix), it is still *similar* to a rotation-scaling matrix (typically called C).

Rotation Scaling Theorem

Let A be a real-valued 2×2 matrix with a complex eigenvalue $\lambda = \alpha + \beta i$, and let $\mathbf{v} = \mathbf{a} + \mathbf{b}i$ be a corresponding eigenvector. Then:

$$A = (\mathbf{a}\ \ \mathbf{b}) \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix} (\mathbf{a}\ \ \mathbf{b})^{-1}$$

If A has complex eigenvalues, then A is similar to a **block-diagonal** matrix C, and we call A **block-diagonalizable**:

Block-Diagonalization of a Matrix

Let A be a real-valued $(m+n) \times (m+n)$ matrix.

- Let $\lambda_1, \lambda_2, \ldots, \lambda_m$ be distinct eigenvalues of A, and let $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m$ be their corresponding eigenvectors.
- Let $\alpha_1 + \beta_1 i, \alpha_2 + \beta_2 i, \ldots, \alpha_n + \beta_n i$ be the corresponding complex eigenvalues of A, and let $\mathbf{a}_1 + \mathbf{b}_1 i, \mathbf{a}_2 + \mathbf{b}_2 i, \ldots, \mathbf{a}_n + \mathbf{b}_n i$ be corresponding eigenvectors.
- Then, A is *similar* to a **block-diagonal** matrix C, where:

$$C = \begin{pmatrix} \lambda_1 & & & & & & & & \\ & \lambda_2 & & & & & & & \\ & & \ddots & & & & & & \\ & & & \lambda_m & & & & & \\ & & & & \alpha_1 & -\beta_1 & & & \\ & & & & \beta_1 & \alpha_1 & & & \\ & & & & & & \ddots & & \\ & & & & & & & \alpha_n & -\beta_n \\ & & & & & & & \beta_n & \alpha_n \end{pmatrix}.$$

- The **block-diagonalization** of A is given by $A = PCP^{-1}$, where:

$$P = \begin{pmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \ldots & \mathbf{v}_m & \mathbf{a}_1 & \mathbf{b}_1 & \mathbf{a}_2 & \mathbf{b}_2 & \ldots & \mathbf{a}_n & \mathbf{b}_n \end{pmatrix}.$$

5.6 Discrete Dynamical Systems

6 Orthogonality and Least Squares

6.1 Inner Product and Orthogonality

Recall that the **dot product** of two vectors in matrix form is $\mathbf{u}^\mathsf{T}\mathbf{v}$, and this is often called the **inner product**.

Orthogonal Subspaces

- **Orthogonal complements** only act on *subspaces*.
- If W is a subspace of $\mathbb{R}^n$, then the **orthogonal complement** of W, denoted $W^\perp$, is the set of all vectors in $\mathbb{R}^n$ that are orthogonal to every vector in W:

$$W^\perp = \left\{\mathbf{v} \in \mathbb{R}^n \mid \mathbf{v}^\mathsf{T}\mathbf{w} = 0 \text{ for all } \mathbf{w} \in W\right\}.$$

  - To show that $\mathbf{v} \in W^\perp$, show that $\mathbf{v}^\mathsf{T}\mathbf{w} = 0$ for all $\mathbf{w} \in W$.
  - If S is a spanning set for $W^\perp$, then ($\mathbf{v}^\mathsf{T}\mathbf{s} = 0$ for all $\mathbf{s} \in S$) $\iff$ $\left(\mathbf{v} \in W^\perp\right)$.
- To find $W^\perp$, find the null space of the matrix whose rows are the basis vectors of W:

$$W^\perp = \mathrm{nul}\left(\begin{pmatrix} \mathbf{w}_1^\mathsf{T} \\ \mathbf{w}_2^\mathsf{T} \\ \dots \\ \mathbf{w}_k^\mathsf{T} \end{pmatrix}\right).$$

  In other words, if $W = \mathrm{col}(A)$, then $W^\perp = \mathrm{nul}\left(A^\mathsf{T}\right)$.
  - $\mathrm{nul}(A) = (\mathrm{row}(A))^\perp$
  - $\mathrm{nul}\left(A^\mathsf{T}\right) = (\mathrm{col}(A))^\perp$
- It holds that $\dim(W) + \dim\left(W^\perp\right) = n$, where n is the dimension of the *ambient space* $\mathbb{R}^n$.
  - This has many implications. For example, if W is geometrically a plane in $\mathbb{R}^3$, then $W^\perp$ is geometrically a line in $\mathbb{R}^3$ that is perpendicular to the plane, such that the dimensions of the plane and line add up to 3. That is, only points along a line perpendicular to the plane are orthogonal to every point in the plane.

Proof: if W is a subspace of $\mathbb{R}^n$, then $W^\perp$ is also subspace of $\mathbb{R}^n$

Let W be a subspace of $\mathbb{R}^n$. We want to show that $W^\perp$ is also a subspace of $\mathbb{R}^n$.

1. First, we show that $\mathbf{0} \in W^\perp$. Since $\mathbf{0}^\top \mathbf{w} = 0$ for all $\mathbf{w} \in W$, we have $\mathbf{0} \in W^\perp$.
2. Next, we show that if $\mathbf{u}, \mathbf{v} \in W^\perp$, then $\mathbf{u} + \mathbf{v} \in W^\perp$. Since $\mathbf{u}^\top \mathbf{w} = 0$ and $\mathbf{v}^\top \mathbf{w} = 0$ for all $\mathbf{w} \in W$, we have:
$$(\mathbf{u} + \mathbf{v})^\top \mathbf{w} = \mathbf{u}^\top \mathbf{w} + \mathbf{v}^\top \mathbf{w} = 0 + 0 = 0.$$
Thus, $\mathbf{u} + \mathbf{v} \in W^\perp$.
3. Finally, we show that if $\mathbf{u} \in W^\perp$ and c is a scalar, then $c\mathbf{u} \in W^\perp$. Since $\mathbf{u}^\top \mathbf{w} = 0$ for all $\mathbf{w} \in W$, we have:
$$(c\mathbf{u})^\top \mathbf{w} = c\left(\mathbf{u}^\top \mathbf{w}\right) = c \cdot 0 = 0.$$
Thus, $c\mathbf{u} \in W^\perp$.

Therefore, since $W^\perp$ contains the zero vector and is closed under vector addition and scalar multiplication, we conclude that $W^\perp$ is a subspace of $\mathbb{R}^n$. ∎

6.2 Orthogonal and Orthonormal Sets

- An **orthogonal set** is a set of vectors that are orthogonal to each other.
  - Formally, a set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is an orthogonal set if $\mathbf{v}_i^\top \mathbf{v}_j = 0$ for all $i \neq j$.
  - All orthogonal set of nonzero vectors is linearly independent.

Proof: An orthogonal set of nonzero vectors is linearly independent

Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ be an orthogonal set of nonzero vectors.

For scalars $c_1, c_2, \dots, c_n$, suppose that $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n = \mathbf{0}$.

Then, for each $i = 1, 2, \dots, n$, we have:

$$\begin{aligned}(c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n)^\top \mathbf{v}_i &= c_1\mathbf{v}_1^\top\mathbf{v}_i + c_2\mathbf{v}_2^\top\mathbf{v}_i + \dots + c_n\mathbf{v}_n^\top\mathbf{v}_i \\ &= c_i\mathbf{v}_i^\top\mathbf{v}_i \\ &= c_i \|\mathbf{v}_i\|^2 = 0 \\ &\to c_i = 0 \text{ since all vectors are nonzero.}\end{aligned}$$

So, $c_1 = c_2 = \dots = c_n = 0$, which means that the set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is linearly independent. ∎

- An **orthogonal basis** for a vector space is a basis consisting of orthogonal vectors.
  - If $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ is an orthogonal basis for a vector space V, and $\mathbf{x} \in V$, then:

$$\mathbf{x} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + \dots + c_n\mathbf{b}_n, \text{ where } c_i = \frac{\mathbf{x} \cdot \mathbf{b}_i}{\mathbf{b}_i \cdot \mathbf{b}_i}.$$

- Let L be the line spanned by a nonzero vector $\mathbf{v}$. That is, let $L = \text{span}\{\mathbf{v}\}$. Then any vector $\mathbf{x}$ not on L can be written as the sum of a vector on L and a vector orthogonal to L. In particular, we can write $\mathbf{x} = \mathbf{p} + \mathbf{q}$, where $\mathbf{p} \in L$ and $\mathbf{q} \in L^\perp$.
  - $\mathbf{p}$ is called the **orthogonal projection** of $\mathbf{x}$ onto L, denoted $\hat{\mathbf{x}}$, and is given by:

$$\mathbf{p} = \text{proj}_L(\mathbf{x}) = \hat{\mathbf{x}} = \left(\frac{\mathbf{x} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}\right)\mathbf{v}.$$

  - $\mathbf{q}$ is called the **orthogonal component** of $\mathbf{x}$ with respect to L, and is given by:

$$\mathbf{q} = \mathbf{x} - \hat{\mathbf{x}} = \mathbf{x} - \left(\frac{\mathbf{x} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}\right)\mathbf{v}.$$

  - The form $\mathbf{p} + \mathbf{q}$ is called the **orthogonal decomposition** of $\mathbf{x}$ with respect to L.
  - The magnitude $|\mathbf{q}|$ is the distance from $\mathbf{x}$ to the line L.

Orthonormal Sets

- An **orthonormal set** is an orthogonal set of vectors where each vector has a length of 1.
  - Formally, a set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ is an orthonormal set if:

$$\mathbf{v}_i^\top \mathbf{v}_j = 0 \text{ for all } i \neq j \text{ and}$$
$$\mathbf{v}_i^\top \mathbf{v}_j = 1 \text{ for all } i = j.$$

- If U is a matrix, then $U^\top U = I$ **iff** the columns of U form an orthonormal set.
  - Similarly, $UU^\top = I$ **iff** the rows of U form an orthonormal set.
- If U has orthonormal columns, then the following properties hold:
  - $|U\mathbf{x}| = |\mathbf{x}|$ (transformation by U preserves length)
  - $(U\mathbf{x})^\top(U\mathbf{y}) = \mathbf{x}^\top\mathbf{y}$ (transformation by U preserves inner product)
  - $(U\mathbf{x})^\top(U\mathbf{y}) = 0 \iff \mathbf{x}^\top\mathbf{y} = 0$ (transformation by U preserves orthogonality)
- If a matrix U is square, then the following are equivalent:
  - U has orthonormal columns
  - U has orthonormal rows
  - $U^\top U = UU^\top = I$.
  - U is invertible, and $U^{-1} = U^\top$.
  - U is an **orthogonal matrix**.

6.3 Orthogonal Decomposition

Let W be a subspace of $\mathbb{R}^n$, and let $\mathbf{x} \in \mathbb{R}^n$. Then, we can write $\mathbf{x} = \mathbf{p} + \mathbf{q}$, where $\mathbf{p} \in W$ and $\mathbf{q} \in W^\perp$.

If $\left\{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_k\right\}$ is an orthogonal basis for W, then:

$$\mathbf{p} = \operatorname{proj}_W(\mathbf{x}) = \hat{\mathbf{x}} = \left(\frac{\mathbf{x} \cdot \mathbf{b}_1}{\mathbf{b}_1 \cdot \mathbf{b}_1}\right)\mathbf{b}_1 + \left(\frac{\mathbf{x} \cdot \mathbf{b}_2}{\mathbf{b}_2 \cdot \mathbf{b}_2}\right)\mathbf{b}_2 + \ldots + \left(\frac{\mathbf{x} \cdot \mathbf{b}_k}{\mathbf{b}_k \cdot \mathbf{b}_k}\right)\mathbf{b}_k.$$

- Trivially, $\mathbf{q} = \mathbf{x} - \hat{\mathbf{x}}$.
- $\mathbf{p}$ is called the **orthogonal projection** of $\mathbf{x}$ onto W.
- $\mathbf{q}$ is called the **orthogonal component** of $\mathbf{x}$ with respect to W.
- The form $\mathbf{p} + \mathbf{q}$ is called the **orthogonal decomposition** of $\mathbf{x}$ with respect to W.

If $\left\{H_1, H_2, \ldots, H_p\right\}$ is a partition of the orthogonal basis $\left\{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_k\right\}$, then:

$$\mathbf{p} = \operatorname{proj}_W(\mathbf{x}) = \hat{\mathbf{x}} = \operatorname*{proj}_{\operatorname{span}(H_1)} \mathbf{x} + \operatorname*{proj}_{\operatorname{span}(H_2)} \mathbf{x} + \ldots + \operatorname*{proj}_{\operatorname{span}(H_p)} \mathbf{x}.$$

Best Approximation Theorem

Let W be a subspace of $\mathbb{R}^n$, and let $\mathbf{x} \in \mathbb{R}^n$. Then, the vector $\hat{\mathbf{x}} = \operatorname{proj}_W(\mathbf{x})$ is the closest vector in W to $\mathbf{x}$, in the sense that:

$$|\mathbf{x} - \hat{\mathbf{x}}| \leq |\mathbf{x} - \mathbf{w}| \text{ for all } \mathbf{w} \in W.$$

In which case, the quantity $|\mathbf{x} - \hat{\mathbf{x}}|$ is called the **distance** from $\mathbf{x}$ to the subspace W.

- Realize that if the basis $\left\{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_k\right\}$ is *orthonormal*, then:

$$\mathbf{p} = \operatorname{proj}_W(\mathbf{x}) = \hat{\mathbf{x}} = \left(\mathbf{x} \cdot \mathbf{b}_1\right)\mathbf{b}_1 + \left(\mathbf{x} \cdot \mathbf{b}_2\right)\mathbf{b}_2 + \ldots + \left(\mathbf{x} \cdot \mathbf{b}_k\right)\mathbf{b}_k.$$

If $U = \begin{pmatrix}\mathbf{b}_1 & \mathbf{b}_2 & \ldots & \mathbf{b}_k\end{pmatrix}$, then $\hat{\mathbf{x}} = UU^\top\mathbf{x}$.

Gram-Schmidt Process

To find an orthogonal basis for a subspace W of $\mathbb{R}^n$, we can apply the **Gram-Schmidt process** to any basis of W.

Let $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k\}$ be *any* basis for W. Then, we can construct an orthogonal basis $\{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_k\}$ for W as follows:

$$\begin{aligned}
\mathbf{b}_1 &= \mathbf{w}_1 \\
\mathbf{b}_2 &= \mathbf{w}_2 - \left(\frac{\mathbf{w}_2 \cdot \mathbf{b}_1}{\mathbf{b}_1 \cdot \mathbf{b}_1}\right)\mathbf{b}_1 && = \mathbf{w}_2 - \operatorname*{proj}_{\operatorname{span}\{\mathbf{b}_1\}} \mathbf{w}_2 \\
\mathbf{b}_3 &= \mathbf{w}_3 - \left(\frac{\mathbf{w}_3 \cdot \mathbf{b}_1}{\mathbf{b}_1 \cdot \mathbf{b}_1}\right)\mathbf{b}_1 - \left(\frac{\mathbf{w}_3 \cdot \mathbf{b}_2}{\mathbf{b}_2 \cdot \mathbf{b}_2}\right)\mathbf{b}_2 && = \mathbf{w}_3 - \operatorname*{proj}_{\operatorname{span}\{\mathbf{b}_1,\mathbf{b}_2\}} \mathbf{w}_3 \\
&\vdots && \vdots \\
\mathbf{b}_k &= \mathbf{w}_k - \left(\frac{\mathbf{w}_k \cdot \mathbf{b}_1}{\mathbf{b}_1 \cdot \mathbf{b}_1}\right)\mathbf{b}_1 - \left(\frac{\mathbf{w}_k \cdot \mathbf{b}_2}{\mathbf{b}_2 \cdot \mathbf{b}_2}\right)\mathbf{b}_2 - \dots - \left(\frac{\mathbf{w}_k \cdot \mathbf{b}_{k-1}}{\mathbf{b}_{k-1} \cdot \mathbf{b}_{k-1}}\right)\mathbf{b}_{k-1} && = \mathbf{w}_k - \operatorname*{proj}_{\operatorname{span}\{\mathbf{b}_1,\mathbf{b}_2,\dots,\mathbf{b}_{k-1}\}} \mathbf{w}_k.
\end{aligned}$$

We can add a *normalization step* to find an orthonormal basis $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ for W:

$$\mathbf{u}_i = \frac{\mathbf{b}_i}{|\mathbf{b}_i|} \text{ for all } i = 1, 2, \dots, k.$$

6.4 Least Squares Problems

The equation $A\mathbf{x} = \mathbf{b}$ is only consistent when $\mathbf{b}$ can be made by taking a linear combination of the columns of A, where weights of that linear combination solve to be $\mathbf{x}$. That is, the system is consistent **iff** $\mathbf{b} \in \operatorname{col}(A)$.

6.5 Inner Product Spaces

An **inner product space** is a vector space V along with a binary operator called the **inner product** $\langle\cdot,\cdot\rangle : V \times V \to \mathbb{R}$. The inner product must satisfy the following properties for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and scalar c:

- commutativity: $\langle \mathbf{u}, \mathbf{v}\rangle = \langle \mathbf{v}, \mathbf{u}\rangle$.
- linearity in the first argument: $\langle c\mathbf{u} + \mathbf{w}, \mathbf{v}\rangle = c\langle \mathbf{u}, \mathbf{v}\rangle + \langle \mathbf{w}, \mathbf{v}\rangle$.
- positive-definiteness: $\langle \mathbf{v}, \mathbf{v}\rangle \geq 0$, and $\langle \mathbf{v}, \mathbf{v}\rangle = 0$ iff $\mathbf{v} = \mathbf{0}$.

A **normed vector space** is a vector space V along with a unary operator called the **norm** $\|\cdot\| : V \to \mathbb{R}$ that satisfies the following properties for all $\mathbf{u}, \mathbf{v} \in V$ and scalar c:

- positive-definiteness: $\|\mathbf{v}\| \geq 0$, and $\|\mathbf{v}\| = 0$ iff $\mathbf{v} = \mathbf{0}$.
- homogeneity: $\|c\mathbf{v}\| = |c|\|\mathbf{v}\|$.
- triangle inequality: $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.

All inner product spaces are normed vector spaces. The norm operator of an inner product space is defined as $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v}\rangle}$, called the norm **induced** by the inner product space.

7 Symmetric Matrices and Quadratic Forms

7.1 Introduction to Symmetric Matrices

- A **symmetric matrix** is a matrix which is equal to its transpose:

$$A \text{ is symmetric} \iff A = A^\top.$$

- If A is a symmetric matrix, then the eigenvectors corresponding to the distinct eigenvalues of A form an orthogonal set.

Proof: Distinct eigenvalues of a symmetric matrix form an orthogonal set

Let A be a symmetric matrix, and let λ_1 and λ_2 be distinct eigenvalues of A with corresponding eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$ respectively. Then:

$$\begin{aligned}
\lambda_1\left(\mathbf{v}_1^\top \mathbf{v}_2\right) &= \left(A\mathbf{v}_1\right)^\top \mathbf{v}_2 && \text{by performing the substitution } A\mathbf{v}_1 = \lambda_1\mathbf{v}_1 \\
&= \mathbf{v}_1^\top A^\top \mathbf{v}_2 && \text{by the property of transpose } (AB)^\top = B^\top A^\top \\
&= \mathbf{v}_1^\top A \mathbf{v}_2 && \text{since } A \text{ is symmetric} \\
&= \lambda_2\left(\mathbf{v}_1^\top \mathbf{v}_2\right) && \text{by performing the substitution } A\mathbf{v}_2 = \lambda_2\mathbf{v}_2
\end{aligned}$$

Since $\lambda_1 \neq \lambda_2$, we must have $\mathbf{v}_1^\top \mathbf{v}_2 = 0$, which means that $\mathbf{v}_1$ and $\mathbf{v}_2$ are orthogonal. ∎

- If A is a symmetric *and* diagonalizable $n \times n$ matrix, then there exists an orthogonal basis $\mathcal{B}$ for $\mathbb{R}^n$ where:
  - Eigenvectors corresponding to eigenvalues of multiplicity 1 are in $\mathcal{B}$
  - For each eigenvalue of multiplicity $m > 1$, we can take m linearly independent eigenvectors corresponding to that eigenvalue, and apply the Gram-Schmidt process to those m eigenvectors to get m orthogonal eigenvectors corresponding to that eigenvalue, which are in $\mathcal{B}$.
  - That is, the eigenbases correpsonding to the distinct eigenvalues of A are *mutually orthogonal*.
- A is **orthogonally diagonalizable** iff there exists a diagonal matrix D and an orthogonal matrix U such that $A = UDU^\top$.
  - To find an orthogonal diagonalization for A, use the Gram-Schmidt process to find an orthonormal basis for every eigenbasis corresponding to an eigenvalue in A with dimension > 1.

Spectral Theorem

If A is a symmetric $n \times n$ matrix:

1. A has n real eigenvalues, counting multiplicities
2. The dimension of each eigenspace will equal the multiplicity of the corresponding eigenvalue.
3. The eigenspaces of A are *mutually orthogonal*.
4. A is orthogonally diagonalizable.

In which case, the decomposition:

$$A = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^\mathsf{T} + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^\mathsf{T} + \ldots + \lambda_n \mathbf{u}_n \mathbf{u}_n^\mathsf{T}$$

is called the **spectral decomposition** of A.

7.2 Quadratic Form

- A **quadratic form** over a set of variables $\{x_1, x_2, \ldots, x_n\}$ is a polynomial over $\{x_1, x_2, \ldots, x_n\}$ in which every term has degree 2:

$$Q(x_1, x_2, \ldots, x_n) = \sum_{i=1}^{n} \sum_{j=1}^{i} a_{ij} x_i x_j.$$

  In general, a quadratic form is a *homogeneous polynomial* of degree 2.

- Any quadratic form $Q(x_1, x_2, \ldots, x_n)$ can be represented in matrix-vector form as:

$$Q(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x},$$

  where A is a symmetric $n \times n$ matrix whose entries are given by:

$$a_{ij} = a_{ji} = \begin{cases} \frac{a_{ij}}{2} & \text{if } i \neq j \\ a_{ij} & \text{if } i = j. \end{cases}$$

  where a_{ij} is the coefficient of $x_i x_j$ and $\mathbf{x} = \begin{pmatrix} x_1 & x_2 & \ldots & x_n \end{pmatrix}^\top$.

- Realize that if $A = I$, then $Q(\mathbf{x}) = \mathbf{x}^\top \mathbf{x} = |\mathbf{x}|^2 = x_1^2 + x_2^2 + \ldots + x_n^2$.

Example: Represent a Quadratic Form as a Matrix-Vector Product

Let $Q(x, y) = 3x^2 + 4xy + 5y^2$ be a quadratic form in x and y. If $\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}$, then find the A such that $Q(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x}$.

We have $a_{11} = 3, a_{12} = a_{21} = 4/2 = 2, a_{22} = 5$. Thus:

$$A = \begin{pmatrix} 3 & 2 \\ 2 & 5 \end{pmatrix} \quad \to \quad Q(\mathbf{x}) = (x \;\; y) \begin{pmatrix} 3 & 2 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Change of Variables in a Quadratic Form

We can make the symmetric matrix A diagonal by making use of the orthogonal diagonalization of A, which must exist since A is symmetric.

Let $A = UDU^\top$ be an orthogonal diagonalization of A, such that U is an orthogonal matrix and D is a diagonal matrix (i.e. U has orthonormal columns). Then $D = U^\top AU$.

Let $\mathbf{y}$ be the vector of our "new" variables, and define $\mathbf{x} = U\mathbf{y} \to \mathbf{y} = U^{-1}\mathbf{x} = U^\top\mathbf{x}$.

Then, we have:

$$Q_x(\mathbf{x}) = \mathbf{x}^\top A\mathbf{x} = \mathbf{x}^\top\left(UDU^\top\right)\mathbf{x} = \left(U^\top\mathbf{x}\right)^\top D\left(U^\top\mathbf{x}\right) = \mathbf{y}^\top D\mathbf{y}.$$

In which case, the columns of U are the **principal axes** of the quadratic form. Since D is a diagonal matrix, our quadratic form ends up only having y_i^2 terms:

$$Q_y(\mathbf{y}) = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \ldots + \lambda_n y_n^2.$$

Change of Variables for a Quadratic Form

Let $Q(\mathbf{x}) = \mathbf{x}^\top A\mathbf{x}$ be a quadratic form, where A is a symmetric matrix. If $\lambda_1, \lambda_2, \ldots, \lambda_n$ are the eigenvalues of A, then:

$$\mathbf{x}^\top A\mathbf{x} = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \ldots + \lambda_n y_n^2.$$

Realize that the coefficients of Q_y are the eigenvalues of A (i.e. the entries on the main diagonal of D). They are called the **principal values** of the quadratic form, and they determine the shape of the graph of the quadratic form.

For example, if all principal values are positive, then Q will *only* output positive numbers. If all principal values are negative, then Q will *only* output negative numbers.

In general, if all eigenvalues are positive, then our quadratic form is **positive definite**, and so is our symmetric matrix A. If all eigenvalues are negative, then our quadratic form is **negative definite**.

If all eigenvalues are non-negative (i.e. positive or zero), then our quadratic form is **positive semidefinite**, and if all eigenvalues are non-positive (i.e. negative or zero), then our quadratic form is **negative semidefinite**.

If some eigenvalues are positive and some are negative, then our quadratic form is **indefinite**.

7.3 Constrained Optimization

A quadratic form $Q(\mathbf{x}) = \mathbf{x}^\top A\mathbf{x}$ subject to the constraint $\mathbf{x}^\top\mathbf{x} = \|\mathbf{x}\|^2 = 1$ attains a **maximum** of its maximum eigenvalue λ_1 when $\mathbf{x} = \mathbf{u}$, where $\mathbf{u}$ is a corresponding unit eigenvector of λ_1. Similarly, Q attains a **minimum** of its minimum eigenvalue λ_p when $\mathbf{x} = \mathbf{v}$, where $\mathbf{v}$ is a corresponding unit eigenvector of λ_p.

Similarly, if λ_k is the k^{th} greatest eigenvalue of A with $\mathbf{u}_k$ a corresponding unit eigenvector, then for all k, λ_k is the maximum value of $Q(\mathbf{x})$ when subject to the constraints:

$$\mathbf{x}^\top\mathbf{x} = 1 \qquad \mathbf{x}^\top\mathbf{u}_1 = 0 \qquad \mathbf{x}^\top\mathbf{u}_2 = 0 \qquad \dots \qquad \mathbf{x}^\top\mathbf{u}_{k-1} = 0.$$

7.4 Singular Value Decomposition

The **singular values** of a matrix A are the square roots of the eigenvalues of $A^\top A$. The singular values of A are always non-negative, and they are typically arranged in non-increasing order: $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_p \geq 0$, where $p = \min(m, n)$. It holds that for all σ_i, σ_i^2 is an eigenvalue of $A^\top A$, so $\sigma_i = \sqrt{\lambda_i}$ if λ_i is an eigenvalue of $A^\top A$.

If A is an $m \times n$ matrix, then there exists an $m \times m$ orthogonal matrix U, an $n \times n$ orthogonal matrix V, and an $m \times n$ diagonal matrix Σ such that:

$$A = U\Sigma V^\top.$$

This is called the **singular value decomposition** of A.

- U is the matrix whose columns are eigenvectors of $A^\top A$.
- V is the matrix whose columns are eigenvectors of $AA^\top$.

A **truncated** singular value decomposition of A is given by:

$$A_k = \sigma_1\mathbf{u}_1\mathbf{v}_1^\top + \sigma_2\mathbf{u}_2\mathbf{v}_2^\top + \dots + \sigma_k\mathbf{u}_k\mathbf{v}_k^\top,$$

where $k < p$ and $\mathbf{u}_i$ and $\mathbf{v}_i$ are the i^{th} columns of U and V respectively. The truncated singular value decomposition of A is the best rank-k approximation of A, in the sense that:

$$\left|A - A_k\right| \leq |A - B| \text{ for all } \operatorname{rank}(B) \leq k.$$